

Course Title:

Financial

Management-I

INTRODUCTION:

MEANING OF FINANCIAL MANAGEMENT

- ❑ **Financial Management:** Is nothing but management of the limited financial resources the organization has to its utmost advantage.
- ❑ It refers to that part of management activity, which is concerned with the *planning* and *controlling* of firm's financial resources.
- ❑ **Financial Management:** is a combination of *two disciplines*, *management and finance*.

❑ **Management**: Is defined as the acts of getting people together to accomplish desired **goals and objectives** of any business areas and organizational activities.

❑ I.e. **Management** consists of the activities of **planning, organizing, staffing, leading, or directing, and controlling** an organization (a group of one or more people or entities) for the purpose of accomplishing a goal or objectives using the **available resources**.

❑ Finance: is defined as the art and science of managing money.

❑ **Finance** is concerned with the process, institutions, markets, and instruments involved in the transfer of money among individuals, businesses, and governments.

Finance as an area of study

❑ **Finance**, in general, consists of **three** interrelated areas:

I. Money and capital markets: which deal with securities markets and financial institutions.

❑ Factors that cause **interest rates** to **rise** and **fall**, the **regulations** to which financial institutions are subjected, and the various types of **financial instruments** such as **bonds, shares, mortgages, certificates of deposits**, and so on.

II. Investments: Which focus on the decision of investors, both individuals and institutions.

□ The **three** major functions in the investment area are:

- (i) Sales of securities,
- (ii) Analysis of individual securities, and
- (iii) Determination of the optimal (best) mix of securities for a given investor.

III. Financial management, (or "*business finance*", "*corporate finance*", or "*managerial finance*"): involves the actual management of business firms.

□ **FM** is concerned with the creation, maintenance and maximization of economic value or wealth through the application of accounting theories and concepts in management decision making.

❑ The **decisions involved are financial decisions** such as:

- ❖ when to introduce a new product,
- ❖ when to invest in new assets,
- ❖ when to replace existing assets
- ❖ when to borrow from banks,
- ❖ when to issue stocks or bonds,
- ❖ when to extend credit to a customer and how much cash to maintain.

❑ **Financial management** is the broadest area in finance and it is important in all types of businesses.

Financial services vs Financial Management:

- i. **Financial services:** is concerned with the design and delivery of advice and financial products (assets) to individuals, businesses and governments within the areas of banking and related institutions, personal financial planning, investments, insurance and so on.

ii. Financial management (managerial finance/corporate finance): is the management of **capital sources** and **their uses** so as to attain the desired goal and objectives of the firm.

❖ It involves **sourcing of funds, making appropriate investments and promulgating the best mix of financial resources** in relation to the **value of the firm.**

KEY ACTIVITIES OF THE FINANCIAL MANAGER

The primary activities of a financial manager are:

1. Performing financial analysis and planning:

❖ **Financial planning (forecasting):** Evaluate productive capacity and determine financing requirement.

✓ **Techniques (Financial ratios) and Interpretations**

❑ The basis to calculate ratios are **historical financial statements.**

❑ However, using the historical data and equipped with the knowledge of future plans and constraints to achieve, management is in a better position to predict future happening to a certain reasonable extent.

❖ **Financial analysis:** Is the process of identifying the **financial strengths and weaknesses** of the firm, by properly establishing the **relationships** between the **items** contained in **balance sheet** and **profit and loss account**.

□ **Ratios** are the symptoms of health of an organization like blood pressure, pulse or temperature of an individual.

❑ Transforming financial data into usable form to monitor financial condition (Increase or Decrease capacity; additional funds or reduction of funds).

2. Making investment and financing decisions:

- **Investment decisions:** Short-term and long-term investments (*Capital budgeting and measurement of expected rate of return*).
- **Financing decisions** "Capital structure and Financing policy (Financial leverage and credit policy plus retention).

3. Managing financial resources:

- ❖ **Management of Current Assets:** (management of working capital, cash, receivable, inventory).
- ❖ **Multi-national financial management considerations**
(Currency Valuation).

□ Financial managers also have the responsibility for deciding:

- The credit terms under **which customers** may buy,
- How much **inventory the company should carry**,
- How much **cash to keep on hand**,
- Whether to acquire other company (merger analysis), and
- How much of the firm's **earnings to retain** in the business versus **payout as dividends**.

FUNCTIONS OF FINANCIAL MANAGEMENT

□ **Financial management** is concerned with **two** distinct **functions**. These are:

1. **Financing function:-** Describes the management of the *sources of capital*
2. **Investing function:-** Concentrates on the **type, size, and** percentage composition of *capital uses*.

SCOPE OF FINANCIAL MANAGEMENT:-

IN TERMS OF APPROACH

- ❑ The **scope** and **complexity** of financial management has been **widening, with the growth of business** in different diverse directions.
- ❑ It has undergone significant changes, over the years in its scope and coverage:

Approaches: Broadly, it has two approaches:

- **Traditional Approach-Procurement of Funds**
- **Modern Approach-Effective Utilization of Funds**

1) Traditional Approach:- Procurement of funds

- ❑ The scope of finance function was **treated in the narrow sense** of procurement or arrangement of funds.
- ❑ The **finance manager** was **treated** as just **provider of funds**, when organization was in need of them.
- ❑ The **utilization or administration of resources** was **considered outside the purview of the finance function** (Finance Manager).

❑ As per this approach, the following aspects only were included in the scope of financial management:

- i. Estimation of required finance,
- ii. Arrangement of funds from financial institutions,
- iii. Arrangement of funds through financial instruments such as shares, debentures, bonds and loans, and
- iv. Looking after the accounting and legal work connected with the raising of funds.

Limitations of Traditional Approach:

- No Involvement of financial manager in Allocation of Funds.
- Financial manager was not Associated in Application of Funds.
- No Involvement of financial manager in day to day Management of funds.

2. Modern Approach:- Effective utilization of funds

- ❑ Since 1950s, the emphasis of Financial Management has been shifted from raising of funds to the effective and judicious (careful) utilization of funds.
- ❑ Financial management is considered as vital and an integral part of overall management.
- ❑ The modern approach is analytical, logical and systematic way of looking into the financial problems of the firm.

- ❑ Advice of finance manager is required at every moment, whenever any decision with involvement of funds is taken.
- ❑ Nowadays, the finance manager is required to look into the financial implications of every decision to be taken by the firm.
- ❑ The involvement of finance manager has been; before taking the decision, during its review and, finally, when the final outcome is judged.

SCOPE OF FINANCIAL MANAGEMENT:-

IN TERMS OF PERSPECTIVE (VIEWPOINT)

I. Balance sheet perspective (how large should an enterprise be?)

- **Uses of funds** (in what form should it hold its Assets?)
- **Sources of funds** (what should be the composition of its claims?)

II. **Income statement perspective** (how fast should it grow?)

- **Sustainable sales growth rate** with out increasing leverage or issuing new shares.
- **Target capital structure:** is the mix of liability and equity (leverage, retention and dividend policy)
- **Capital intensity:** Sales per invested capital, Investment turnover (return).

OBJECTIVES OF FINANCIAL MANAGEMENT

❑ The term **objective** is used in the sense of a **goal or decision criterion** for the **four decisions**:

- ❖ **Investment decision**: deal with allocation of the firm's scarce financial resources among competing uses.
- ❖ **Dividend policy decision**: address the question how much of the cash a firm generates from operations should be distributed to owners in the form of dividends and how much should be retained by the business for further expansion.
- ❖ **Financial decision**: deal with the financing of the firm's investments, i.e., decisions whether the firm should use equity or debt funds in order to finance its assets.

❖ Asset management decision

- ***Determining the asset mix or composition:*** - determining the total amount of the firm's finance to be invested in current and fixed assets.
- ***Determining the asset type:*** - determining which specific assets to maintain within the categories of current and fixed assets.
- ***Managing the asset structure,*** i.e., maintaining the composition of current and fixed assets and the type of specific assets under each category

❑ The financial manager uses the overall company's **goal** of *shareholders' wealth maximization* which is reflected through:

- The **increased dividend per share**, and
- The **appreciations of the prices of shares**

THE FOLLOWING ARE **SPECIFIC OBJECTIVES** OF FINANCIAL MANAGEMENT:

I. Determining the Size and Growth Rate of firm:

- The size of the firm is equal to the total assets as indicated in the balance sheet and measured by the yearly percentage change.

II. Determining Assets Composition (Portfolio):

- Real assets (Current assets and fixed assets) Or Financial assets (Bonds, stocks, loans, advances, and negotiable securities....etc)
- ❑ The percentage composition of the assets of the firm is computed as ratio of the book value of each asset to total book values of all assets.

❑ The choice of the percentage composition of asset items affects the level of business risk.

❑ The wealth maximizing assets structure can be described in either of the following ways:

- ✓ The *asset structure that yields the large profit for a given level of exposure to business risk,* or
- ✓ The *asset structure that minimizes exposure to business risk that is needed to generate the desired profit.*

III. Determining the Composition of Liabilities and Equity:

- ❑ liabilities and equity are the **sources of capital** for business firms.
- ❑ The **mix of liabilities and equity** of the business is what is known as the **capital structure**.
- ❑ The **liability and equity percentage compositions** of the business firm is measured by **dividing the book value of each liability or equity item** by the **total book values of all liabilities and equity**.

□ When the business firm **finances its investment by using debt capital**, (*"leverage" being the jargon used in Finance to refer this*), the business firm and its shareholders face **added risks** along with the possibility of **added returns**.

➤ The **added risk** is the possibility that the firm may face **difficulty to repay its debts** as they mature. (*This is referred to as a negative leverage*)

➤ The **added returns** come from the ability of the firm to **earn the rate of return higher than the interest payments and related financing costs** of using liabilities. (*This is referred to as a positive leverage*)

❑ The added returns may be paid as dividends and/or re-invested in the firm to generate more profit.

❑ This, in effect, would **maximize the wealth of shareholders of the business firm.**

GOAL OF THE FIRM: Profit vs Wealth Maximization

- ❑ The **business decisions** that financial managers are required to make **entirely depend on the purposes or goals** of their respective organizations.
- ❑ So, it is very important to distinguish between *wealth maximization and profit maximization* as goals of business firms.

1. Profit Maximization:

- ❑ Profit-maximization is a **traditional micro-economics theory** of business firm, which was historically considered as the goal of the firm.
- ❑ It stresses on the **efficient use of financial/capital resources** of the firm.
- ❑ **However**, as a goal of the business firm **ignores** many of the **real world complexities** that financial managers try to address in their decisions.

✓ **Profit maximization** looks at the total company profit rather than profit per share.

✓ **Profit maximization** does not speak about the company's dividends as either a return to shareholders or the impact of dividend policy on stock prices.

□ In the more applied discipline of financial management, however, firms must deal every day with two major factors: These are uncertainty and timing.

A. Uncertainty of Returns/Risk

❑ Profit maximization as the goal of business firms ignores uncertainty and risks.

✓ Projects and investment alternatives are compared by examining their expected values or weighted average profits.

✓ Whether or not one project is riskier than another, doesn't enter these calculations; economists do discuss risk, but tangentially (or imaginatively).

❑ **In reality**, however, projects differ in a great deal with respect to the risk characteristics, and ignoring these differences can result in incorrect decisions.

B. Timing of Returns

- ❑ Another problem with profit maximization as the goal of business firm is that it ignores the timing of the returns from projects.
- ❑ From the concept of "Time Value of Money ", money has a definite time value as people have definite preference for current benefits over future benefits.

- ❑ The returns obtained can be re-invested at the prevailing rate of return.
- ❑ The financial manager must always consider the possible timing of returns (profits) in financial decision-making.
- ❖ These limitations of profit maximization as the goal of business firms lead us to the maximization of the more robust goal of the business firm, that is, shareholders' wealth.

2. Wealth Maximization

- ❑ Wealth maximization, on the other hand, is a more *comprehensive model dealing* with the goal of the firm.
- ❑ According to this model, it is made clear that there are two ways in which the wealth of shareholders changes. These are:
 - Through *changing dividend payments*, and
 - Through *the change in the market price of common shares*

□ Hence, the change in shareholders' wealth, or change in the value of business firms, may be calculated as follows:

- 1) Multiply the dividend per share paid during the period by the number of shares owned.
- 2) Multiply the change in shares price during the period by the number of shares owned.
- 3) Add the dividends and the change in the market value of shares, computed in step 1 and 2 above, to obtain the change in the shareholders' wealth during the period.

❑ In order to maximize the wealth of shareholders, a business firm must seek to provide the largest attainable combination of **dividends per share and stock price appreciation**.

THE AGENCY PROBLEM

- ❑ While the goal of the business firm is the maximization of shareholders' wealth, in reality the agency problem may interfere with the implementation of this goal.
- ❑ The agency problem is the result of a separation of the management and the ownership of the firm.

❑ Because of the separation between the decision makers and owners, managers may make decisions that are not in line with the goal of the business firm, or not consistent with the interests of owners, that is outlined as maximization of shareholders' wealth.

END OF CHAPTER 1

THANK YOU!

CHAPTER 2

FINANCIAL ANALYSIS AND PLANNING

Introduction

- Financial analysis is a tool of financial management.
- It consists of the evaluation of the **financial condition** and **operating results** of a business firm and the forecasting of its future condition and performance.

Sources of Financial Information

1. The Balance Sheet

- The balance sheet shows the *financial position* of a firm at a particular point of time.
- A completed balance sheet shows information such as the total **value of assets**, total *indebtedness*, *equity*, available cash and value of *liquid assets*.

Sources of Financial Information...

2. Income Statement

- An income statement often provides a better measure of the operation's performance and profitability.
- It shows the operating results of a firm, flows of revenue and expenses.

Sources of Financial Information...

3. Cash Flow Statement

- Reports the sources and uses of the operation's cash resources.
- Such statements **not only show** the change in the operation's cash resources throughout the year, but also when the *cash was received or spent*.

Financial Analysis

- Financial analysis is the assessment of **firm's past**, **present**, and **anticipated future financial condition**.
- Its objectives are to determine the *firm's financial strength* and to **identify its weaknesses**.

Financial Analysis...

The Need for Financial Analysis

- The following stakeholders are interested in financial statement analysis to make their respective decision at right time.

- 1. Investors**
- 2. Lenders**
- 3. The Management**
- 4. Suppliers**
- 5. Employees**
- 6. Government bodies**
- 7. Competitors**

Methods/ Domains of Financial Analysis

RATIO ANALYSIS

- The most widely used financial analysis technique is ratio analysis, the analysis of relationships between two or more line items on the financial statements.
- **A ratio:** Is the mathematical relationship between two quantities in the financial Statement.

Benchmarks for Evaluation

1. Rule of thumb:

- Has the virtue of simplicity but has little to recommend it conceptually.

3. Time-Series Analysis:

- Is applied when a financial analyst evaluates performance of a firm over time.

2. Cross-Sectional Analysis:

- ❖ Involves the comparison of different firm's financial ratios at the same point in time.

Types of Financial Ratios

- There are **five basic** categories of financial ratios.
- The categories consist of **liquidity, activity, leverage, profitability and market value ratios.**

Each category is explained by using an example set of financial ratios for **Merob Company**.

Merob Company, Income Statements		
Variables	2001	2000
Sales	3,074,000	2,567,000
Less Cost of Goods Sold	2,088,000	1,711,000
Gross Profit	986,000	856,000
Less Operating Expenses		
Selling Expenses	100,000	108,000
General and Adm. Expenses	468,000	445,000
Total Operating Expenses	568,000	553,000
Operating Profit	418,000	303,000
Less Interest Expenses	93,000	91,000
Net Profit Before Tax	325,000	212,000
Less Profit Tax (at 29%)	94,250	61,480
Net Income After Tax	230,750	150,520
Less Preferred Stock Dividends	10,000	10,000
Earning Available to Common	220,750	140,520
Shareholders		

Merob, Balance Sheets		
	2001	2000
Assets		
Current Assets		
Cash	363,000	288,000
Marketable Securities	68,000	51,000
Accounts Receivables	503,000	365,000
Inventories	289,000	300,000
Total Current Assets	1,223,000	1,004,000
Gross Fixed Assets (at cost)		
Land and Buildings	2,072,000	1,903,000
Machinery and Equipment	1,866,000	1,693,000
Furniture and Fixture	358,000	316,000
Vehicles	275,000	314,000
Others	98,000	96,000
Total Fixed Assets	4,669,000	4,322,000
Less Acc. Depreciation	2,295,000	2,056,000
Net Fixed Assets	2,374,000	2,266,000
Total Assets	3,597,000	3,270,000
Liabilities and Owners' Equity		
Current Liabilities		
Accounts Payable	382,000	270,000
Notes Payable	79,000	99,000
Accruals	159,000	114,000

1. Liquidity Ratios

- Liquidity refers to, the ability of a firm to meet its **short-term financial obligations** when and as they fall due.
- Liquidity ratios provide the basis for answering the questions: Does the firm have **sufficient cash** and **near cash assets** to pay its bills on time?

Liquidity Ratios...

A. Current Ratio: - Measures a firm's ability to satisfy or cover the claims of short term creditors by using only current assets.

That is, it measures a firm's short-term solvency or liquidity.

The current ratio is calculated by dividing current assets to current liabilities

$$\text{Current Ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

Liquidity Ratios...

Therefore, the current ratio for Merob Company is

$$\text{For 2001} = \frac{1,223,000}{620,000} = \underline{\underline{1.97}}$$

- The unit of measurement is either birr or times.
- So, we could say that Merob has Birr 1.97 in current assets for every 1 birr in current liabilities, or, we could say that Merob has its current liabilities covered 1.97 times over.
- An ideal current ratio is 2:1 or more.

Liquidity Ratios...

B. Quick (Acid-test) Ratio: This ratio measures the short term liquidity by removing the least liquid assets such as:

Inventories: are excluded because they are not easily and readily convertible into cash and more over, losses are most likely to occur in the event of selling inventories.

Liquidity Ratios...

- **Prepaid Expenses** such as; prepaid rent, prepaid insurance, and prepaid advertising, pre paid supplies are excluded because they are not available to pay off current debts.

$$\text{Quick/Acid test Ratio} = \frac{\text{current asset} - \text{inventory}}{\text{current liability}}$$

- For 2001, Quick Ration for Merob Company will be:

$$= \frac{1,223,000 - 289,000}{620,000} = \underline{\underline{1.51}}$$

- Interpretation: Merob has 1birr and 51 cents in quick assets for every birr current liabilities.
- A moderately high ratio is required by the firm.

2. Activity Ratios

- Activity ratios are also known as **assets management or turnover ratios**.
- These ratios indicate how well the firm manages its assets.
- They provide the basis for assessing how the firm is **efficiently or intensively using its assets to generate sales**.
- These ratios are called **turnover ratios** because they show the speed with which assets are being converted into sales.

Activity Ratios

- Let us see the case of ABC and XYZ café having different compositions of current assets but equal in total amount.

Exercise 2.2 Café	ABC Café	XYZ
	(Birr)	(Birr)
Cash	0	7,000
Marketable Security	0	17,000
A/R	0	5,000
Inventories	<u>35,000</u>	<u>6,000</u>
Total Current Asset	<u>35,000</u>	<u>35,000</u>
Current Liabilities	0	6,000
A/P	14,000	2,000
N/P	0	4,000
Accruals	<u>0</u>	<u>2,000</u>
Total Current Liability	<u>14,000</u>	<u>14,000</u>

Activity Ratios

$$\begin{aligned} \text{CR} &= \frac{\text{CA}}{\text{CL}} & \text{CR} &= \frac{\text{CA}}{\text{CL}} \\ & & &= \frac{35,000}{14,000} = 2.5 \text{ times} \\ &= \frac{35,000}{14,000} = \underline{\underline{2.5}} \text{ times} \end{aligned}$$

- This differences cause the **activity ratio** showing the composition of each assets than the current ratio making activity ratio more important in showing a 'true' liquidity position than current ratio.
- High turnover ratios usually associated with good assets management and lower turnover ratios with poor assets management.

Activity Ratios...

1. Inventory Turnover Ratio

- It is a rough measure of how many times per year the inventory level is **replaced or turned over.**

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventories}}$$

Activity Ratios...

- For Merob Company Inventory Turnover for 2001=
 $2,088,000 / 294,500 * = \underline{\underline{7.09}}$ times
- Interpretation: - Merob's inventory is sold out or turned over 7.09 times per year.
- In general, a high inventory turnover ratio is better than a low ratio.

Activity Ratios...

2. Average Age of Inventory

- The number of days inventory is kept before it is sold to customers. It is calculated by dividing the number of days in the year to the inventory turnover.

$$\text{Average Age of Inventory} = \frac{\text{number of days in a year} / 365 \text{ days}}{\text{inventory turn over}}$$

- Therefore, the Average Age of Inventory for Merob Company for the year 2001 is:

$$\frac{365 \text{ days}}{7.09} = \underline{\underline{51}} \text{ days}$$

7.09

- This tells us that, roughly speaking, inventory remain in stock for 51 days on average before it is sold.

Activity Ratios...

3. Accounts Receivable Turnover Ratio

That is, it indicates how many times or how rapidly accounts *receivable is converted into cash during a year.*

- The accounts receivable turnover is a comparison of the size of the company's sales and its uncollected bills from customers.

Activity Ratios...

$$\text{Receivable Turnover} = \frac{\text{Net Sales}}{\text{Average Account Receivables}}$$

The accounts receivable turnover for Merob Company for the year 2001 is computed as under.

$$\text{Accounts receivable turnover ratio} = \frac{3,074,000}{434,000*} = \underline{\underline{7.08}}$$

Interpretation: Merob Company collected its outstanding credit accounts and re-loaned the money 7.08 times during the year. Reasonably high accounts receivable turnover is preferable.

Activity Ratios...

4. Average Collection Period: Shows how long it takes for account receivables to be cleared (collected). The average collection period represents the number of days for which credit sales are locked in with debtors (accounts receivables).

- Assuming 365 days in a year, average collection period is calculated as follows.

Average Collection period = $365/\text{Receivable Turnover}$

- The average Collection period for Merob Company for the year 2001 will be: $365 \text{ days}/7.08 = 51 \text{ days}$ or **Average collection period = Average accounts receivables* 365 days/Sales**

$$434,000 * 365 \text{ days} / 3,074,000 * = \underline{\underline{51}} \text{ days}$$

The **higher average** collection period is an indication of **reluctant collection policy** where much of the firm's cash is tied up in the form of accounts receivables, whereas, **the lower the average** collection period than the standard is also an **indication of very aggressive collection policy** which could result in the reduction of sales revenue.

Activity Ratios...

5. Average Payment Period

- The average Payment Period/ Average Age of accounts Payable shows, the time it takes *to pay to its suppliers*.

The Average Payment Period =
$$\frac{\text{Accounts Payable}}{\text{Average purchase per day}}$$

Purchase is estimated as a given percentage of cost of goods sold. Assume purchases were 70% of the cost of goods sold in 2001.

So, Average Payment Period =
$$\frac{382,000}{2,088,000 \times .70 / 365} = \underline{95} \text{ days}$$

- This shows that, on average, the firm pays its suppliers in 95 days.
- The longer these days, the more the *credit financing* the firm obtains from its suppliers.

Activity Ratios...

6. Fixed Asset Turnover: Measures the efficiency with which the firm has been using its fixed assets to generate revenue.

- The Fixed Assets Turnover for Merob Company for the year 2001 is calculated as follows.

Fixed asset turnover = Net sales

Net Fixed Asset

$$\text{Fixed Assets Turnover} = \frac{3,074,000}{2,374,000} = \underline{\underline{1.29}}$$

This means that, Merob Company has generated birr 1.29 in net sales for every birr invested in fixed assets.

Activity Ratios...

7. Total Asset Turnover- Measures a firm's efficiency in management its total assets to generate sales.

$$\text{Total Assets Turnover} = \frac{\text{Net sales}}{\text{Net total assets}}$$

The Total Assets Turnover for Merob Company for the year 2001 is as follows. $\frac{3,074,000}{3,597,000} = 0.85$

Interpretation:- Merob Company generates birr 0.85 (85 cents) in net sales for every birr invested in total assets.

- A high ratio suggests greater efficiency in using assets to produce sales where as, a low ratio suggests that Merob is not generating a sufficient volume of sales for the size of its investment in assets.

3.LEVERAGE RATIOS

Leverage ratios are also called solvency ratio. **Solvency is a firm's ability to pay long term debt** as they come due. There are two types of debt measurement tools. These are:

A. Financial Leverage Ratio: These ratios examine **balance sheet ratios** and determine the extent to which borrowed funds have been used to finance the firm. It is the relationship of borrowed funds and owner capital.

B. Coverage Ratio: These ratios measure the **risk of debt and calculated** by income statement ratios designed to determine the number of times fixed charges are covered by operating profits.

Leverage Ratios...

A.1, Debt Ratio: Shows the percentage of assets financed through debt. It is calculated as:

$$\text{Debt ratio} = \frac{\text{Total Liability}}{\text{Total Assets}}$$

The debt ratio for Merob Company for the year 2001 is as follows: $= \frac{1,643,000}{3,597,000} = 0.457$ or 45.7 %

- This indicates that the firm has financed 45.7 % of its assets with debt. Higher ratio shows more of a firm's assets are provided by **creditors** relative to owners.

Leverage Ratios...

A.2. Debt -Equity Ratio: express the relationship between the amount of a firm's total assets financed by creditors (debt) and owners (equity). Thus, this ratio reflects the relative claims of creditors and shareholders' against the asset of the firm.

$$\text{Debt-equity ratio} = \frac{\text{Total Liability}}{\text{Stockholders' Equity}}$$

The Debt- Equity Ratio for Merob Company for the year 2001 is indicated as follows.

$$\text{Debt- Equity ratio} = \frac{1,643,000}{1,954,000} = \mathbf{0.84 \text{ or } \underline{\underline{84\%}}}$$

Interpretation: lenders' contribution is 0.84 times of stock holders' contributions.

Leverage Ratios...

B. 1. Times Interest Earned Ratio: Measures the ability of a firm to pay interest on a timely basis.

$$\text{Times Interest Earned Ratio} = \frac{\text{Earning Before Interest and Tax}}{\text{Interest Expense}}$$

The times interest earned ratio for Merob Company for the year 2001 is: $\frac{418,000}{93,000} = \underline{\underline{4.5}}$ times

- This ratio shows the fact that earnings of Merob Company can decline 4.5 times without causing financial losses to the Company, and creating an inability to meet the interest cost.

Leverage Ratios...

B.2.Coverage Ratio: The problem with the times interest earned ratio is that, it is based on earning before interest and tax, which is not really a measure of cash available to pay interest. One major reason is that, depreciation, a non cash expense has been deducted from earning before Interest and Tax (EBIT). Since interest is a cash outflow, one way to define the cash coverage ratio is as follows:

$$\text{Cash Coverage Ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest}}$$

(Depreciation for 2000 and 2001 is 223,000 and 239,000 respectively).

So, Cash coverage ratio for Merob Company for the year 2001 is $\frac{418,000 + 239,000}{93,000} = \underline{7.07}$ times

- This ratio indicates the extent to which earnings may fall without causing any problem to the firm regarding the payment of the interest charges.

4. Profitability Ratios

- **Profitability** is the ability of a business to earn profit over a period of time.
- Creditors want to get interest and repayment of principal regularly. Owners want to get a required rate of return on their investment. These ratios include:
 - I. Gross Profit Margin
 - II. Operating Profit Margin
 - III. Net Profit Margin
 - IV. Return on Investment
 - V. Return on Equity
 - VI. Earning Per Share

Profitability Ratios...

I. Gross Profit Margin: This ratio computes the margin earned by the firm after incurring manufacturing or purchasing costs. It indicates management effectiveness in pricing policy, generating sales and controlling production costs. It is calculated as:

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$

The gross profit margin for Merob Company for the year 2001 is:

$$\text{Gross Profit Margin} = \frac{986,000}{3,074,000} = \underline{\underline{32.08\%}}$$

Interpretation: Merob company profit is 32 cents for each birr of sales.

- A **high gross profit margin ratio** is a sign of good management.

Profitability Ratios...

II. Operating Profit Margin: This ratio is calculated by dividing the net operating profits by net sales. The net operating profit is obtained by deducting depreciation from the gross operating profit. The operating profit is calculated as:

$$\text{Operating Profit Margin} = \frac{\text{Operating Profit}}{\text{Net Sales}}$$

The operating profit margin of Merob Company for the year 2001 is: $\frac{418,000}{3,074,000} = \underline{\underline{13.60}}$ •

- Interpretation: Merob Company generates around 14 cents operating profit for each of birr sales.

Profitability Ratios...

III. Net Profit Margin: is obtained by subtracting operating expenses and income taxes from the gross profit. This ratio measures the ability of the firm to turn each birr of sales in to net profit. It is calculated as:

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

The net profit margin for Merob Company for the year 2001 is:

$$\frac{230,750}{3,074,000} = \underline{\underline{7.5 \%}}$$

- This means that Merob Company has acquired 7.5 cents profit from each birr of sales.

Profitability Ratios...

IV. Return on Investment (ROI): The return on investment also referred to as Return on Assets measures the overall effectiveness of management in generating profit with its available assets, i.e. how profitably the firm has used its assets.

The return on assets is calculated as:

$$\text{Return on Assets (ROA)} = \frac{\text{Net Income}}{\text{Total Assets}}$$

return on assets for Merob Company for the year 2001 is: •

$$\frac{230,750}{3,597,000} = \underline{\underline{6.4 \%}}$$

Interpretation: Merob Company generates little more than 6 cents for every birr invested in assets.

Profitability Ratios...

V. Return on Equity: The shareholders of a company may Comprise Equity share and preferred share holders.

- Preferred shareholders are the shareholders who have a priority in receiving dividends (and in return of capital at the time of winding up of the Company). The rate of dividend on the preferred shares is fixed.
- The Return on equity is calculated as:

$$\text{ROE} = \frac{\text{Net Income}}{\text{Stockholders Equity}}$$

- The Return on equity of Merob Company for the year 2001 is:

$$\frac{230,750}{1,954,000} = \underline{\underline{11.8\%}}$$

- Interpretation: Merob generates around 12 cents for every birr in shareholders equity.

Profitability Ratios...

VI. Earning per Share (EPS): It reveals the profit available to each ordinary share. It is calculated by dividing the profits available to ordinary shareholders (i.e. profit after tax minus preference dividend) by the number of outstanding equity shares.

The earning per share is calculated as:

EPS = Earning Available for Common Stockholders

Number of Shares of Common Stock Outstanding

- Therefore, the earning per share of Merob Company for the year 2001 is: $EPS = \frac{220,750}{76,262} = \text{birr } 2.90 \text{ per share}$

76,262 shares

- Interpretation: Merob Company earns birr 2.90 for each common shares outstanding.

5. Market Value Ratio

- They measures the performance of the **firm's common stocks in the capital market.**
- This is known as the market value of equity and reflects the risk and return associated with the firm's stocks.
- The following are the important valuation ratios:

Market Value Ratio...

A. Price- Earnings (P/E) Ratio: The price earning ratio is an indicator of the firm's growth prospects, risk characteristics, shareholders orientation corporate reputation, and the firm's level of liquidity. The P/E ratio can be calculated as:

$$\text{P/E Ratio} = \frac{\text{Market Price per Share}}{\text{Earning per Share}}$$

The price per share could be the price of the share on a particular day or the average price for a certain period.

- Assume that Merob Company's common stock at the end of 2001 was selling at birr 32.25, using its EPS of birr 2.90, the P/E ratio at the end of 2001 is:

$$= 32.25 / 2.90 = \underline{\underline{11.10}}$$

- This figure indicates that, investors were paying birr 11.10 for each 1.00 of earnings.

Market Value Ratio...

B. Market Value to Book Value (Market-to-Book) Ratios

- The market value to book value ratio is a measure of the firm's contributing to wealth creation in the society. It is calculated as:

$$\text{Market-Book Ratio} = \frac{\text{Market Value per Share}}{\text{Book Value per Share}}$$

The book value per share can be calculated as:

$$\text{Book value per share} = \frac{\text{Total Stockholders Equity}}{\text{No. of Common Shares Outstanding}}$$

- Book Value per Share for 2001 = $\frac{1,954,000}{76,262} = \underline{\underline{25.62}}$
- Therefore, the Market-Book Ratio for Merob Company for the year 2001 is: $\frac{32.25}{25.62} = \underline{\underline{1.26}}$
- The market –to-book value ratio is a relative measure of how the growth option for a company is being valued vis-à-vis its physical assets. The greater the expected growth and value placed on such, the greater this ratio.

END OF CHAPTER 2

THANK YOU!

Chapter Three

Time value of money (TVM)

5.1. Introduction

- 🍏 Business organizations deals with **interest rates** when it makes both **financing** and **investment** decisions.
- A company can, therefore, **earn a rate of return** on its **invested** funds and **a rate of interest** on the funds it **lent** to borrowers.
- The **key concept** that underlies this is the **time value of money**: that “**a birr today is worth more than a birr received a year from now**”.
- This is **because** if you had it now, you could **invest** that birr or given as **loans** during the year and **earned a return** or **an interest on it**.
- In general business terms, **interest** is defined as **the cost of using money overtime**.

- This definition is in **close agreement** with the definition used by **economists**, who prefer to say that **interest represents** the **time value of money**.
- **Ignoring** the effects of **inflation**, a dollar today is **worth more** than a dollar to be received a year from now.
- **In other words**, we would **all prefer to receive** a specific amount of money **now** rather than on **some future date**.
 - ✓ This **preference rests** on the **time value of money**.
- 🍏 TVM is also described as **discounted cash flow (DCF)**.
 - ✓ DCF is **a technique used** to determine the **present value** of a certain amount of money received at a future date.
- The **interest rate** is used as the **discounting factor**, which can be found by using a present value (PV) table.

- ❑ **Inflows** of dollars on various future dates **should not be added together** as if they were of equal value.
 - These **future cash inflows** must be **restated** at their **present values** before they are aggregated.
 - ➡ The concept of the time value of money tells us that **more distant cash inflows have a smaller present value** than cash inflows to be received within a **shorter time span**.
 - ➡ “The **latter** money is received, the **less value** it holds, and Br. 1 today is **worth more** than Br. 1 received at a date in the future”.
- ❑ Similar reasoning applies to **cash outflows**.
 - ➡ Before we add together cash outflows on various future dates, we must **restate** these outflows at their **present values**.
 - ➡ The **more distant** the date of a cash outflow, the **smaller** is its **present values**.

- As a **simple example** of this **concept of present value**, assume that you are trying **to sell your car** and you **receive offers** from **three** prospective buyers.
- ❖ **Buyers “A” offers you Br. 8000 to be paid immediately. Buyer “B” offers you Br. 8,200 to be paid one year from now. Buyer “C” offers the highest price, Br. 9,200 but this offer provides that payment will be made in five years. Assuming that the offers by “B” and “C” involves no credit risk and that money may be invested at 5% interest compounded annually, which offer would you accept?**
- ➡ **You should accept the offer of Br. 8000 to be **received immediately**, because the **present value** of the **other two offers** is **less than** Br. 8000.**
 - ✓ If you were to invest Br. 8000 today, even at the modest rate of interest of 5%, your investment would be **(8,400)** more than Br. 8200 in one year and considerably **(10,000)** more than Br. 9,200 in five years.

- This **example** suggests that **the timing of cash receipts** and **payments** has an **important effect** on the **economic worth** and the **accounting values** of both assets and liabilities.
- **Consequently**, **investment** and **borrowing decisions** should be made **only after a careful analysis** of the **relative present values** of the prospective cash inflows and outflows.

5.2. Simple Interest and Compound Interest

- **Interest** is the **excess of resources** (usually cash) **received** or **paid** over the amount of resources **loaned** or **borrowed** at an earlier date.
 - **Business transactions** **subject to interest** **state** whether **simple** or **compound** interest is to be calculated.
- 1) **Simple interest** is the **return** on a principal amount for **one time period**.
- We may also think of **simple interest** as a **return** for more than one **time period** if we **assume** that the **interest itself** does not earn a **return**, but this kind of situation occurs rarely in the business world.
 - Simple interest is **usually applicable only** to **short-term investment** and **borrowing transactions** involving a **time span** of less than one **year**.

- Interest is **expressed** in **terms** of **an annual rate**.
- The formula for simple interest is:

$$I = p * r * t$$

(Interest = principal x annual rate of interest x number of years **or** fraction of a year that interest accrues).

Or, using alternative approach,

$$F = P + I$$

Then, substitute $I = P * i * n$ in the expression to obtain

$$F = P + Pin$$

$$F = P (1 + in)$$

For example, interest on Br. 10,000 at 8% for one year is expressed as follows:

$$I = p.r.t$$

$$I = \text{Br. } 10,000 \times 0.08 \times 1$$

$$I = \underline{\text{Br. } 800}$$

- The amount to be **repaid** at the end of the year is the **maturity (future) value** of the specified money.

Accordingly, $F = P + I$

$$F = 10000 + 800$$

$$F = \underline{\text{Br. } 10,800}$$

Exercise:

1. Salon borrowed \$5,000 at per year simple interest of 5% for two years to buy new hair dryers. How much interest must be paid?
2. Marcus Logan can purchase furniture with a two-year simple interest loan at 9% interest per year. What is the maturity value for a \$2,500 loan?

- 2) **Compound interest** is the **return on** a **principal amount** for **two or more time periods**, **assuming** that the **interest** for each time period is **added to** the principal amount **at the end** of each period, and **earns interest** in all subsequent periods.
- Is calculated on the **principal amount** and on the **accumulated interest** of previous period (and therefore be regarded as “**interest on interest**”)
 - **Because** most investment and borrowing transactions involve more than one time period, business executives **evaluate** proposed transactions in **terms of periodic returns**, each of which is **assumed to be reinvested** to **yield additional returns**.

For example, if interest at 8% is compounded **quarterly** for one year on a principal amount of Br. 10,000 the **total interest** (compound interest) would be Br. 824.32, as computed below:

Period	Principal x Rate x Time	= Compound Interest	Accumulated Amounts
1st quarter---	Br. 10,000 x 0.08 x $\frac{1}{4}$	Br. 200	Br. 10,200
2nd quarter-----	10,200 x 0.08 x $\frac{1}{4}$	204	10,404
3rd quarter-----	10,404 x 0.08 x $\frac{1}{4}$	208.08	10,612.08
4th quarter----	10,612.08 x 0.08 x $\frac{1}{4}$	212.24	<u>10,824.32</u>
Interest-----		<u>824.32</u>	

N.B,

- A period, for this purpose, can be any unit of time.
- If interest is compounded annually, a year is the appropriate compounding or conversion or interest period.
- If it is compounded monthly, a month is the appropriate period.
- It is important to know that the number of compounding period/s within a year is/are used in order to find the interest rate per compounding periods and it is denoted by i in the above formula.

- Consequently, when the interest rate is stated as annual interest rate and is compounded more than once a year, the *interest rate per compounding period* is computed by the formula:

$$i = j / m$$

Where; j = is annual quoted or nominal interest rate
 m = number of conversion periods per year or the
compounding periods per year

$$n = m \times t$$

Where; t = is the number of years

- In the computation of compound interest, the accumulated amount (interest) at the end of each period becomes the principal amount for purposes of computing interest for the following period.

- In computing simple interest, the **number** of years or **time**, **n**, can be measured in days.
- **In such case**, there are **two ways** of computing the interest.
- 1) **The Exact Method**: if a year is considered as 365 days, the interest is called **exact simple interest**.
- If the exact method is used to calculate interest, then the **time** is

$$n = \text{number of days}/365$$

- 2) **The Ordinary Method (*Banker's Rule*)**: if a year is considered as 360 days, the interest is called **ordinary simple interest**.
- The time, is calculated as

$$n = \text{number of days}/360$$

5.3 Future and Present Values

- **Future value** involves a **current amount** that is **increased** in the future **as a result of** **compound interest accumulation**.
- **Present value**, in contrast, involves a **future amount** that is **decreased** to the present as **a result of** **compound interest discounting**.
- **Discounting**, in effect, **extracts** the **interest** from a **future value** thereby **returning** to the principal amount.
- The fact that investments have **starting points** and **ending points** makes it **easier to** understand present and future values.
- **Present value** in general refers to **dollar (birr) values** at the **starting point** of an investment, and **future value** refers to **end-point dollar (birr) values**.

- If the dollar (birr) **amount** to be invested **at the start is known**, the **future value** of that amount **at the end can be projected**, provided the **interest rate** and **number** of interest compounding **periods** are also specified.
- **Similarly**, if the dollar (birr) amount available **at the end** of an investment period (future value) **is known**, the amount of **money needed** at the start of the investment period (present value) **can be determined**, again if the **interest rate** and **number** of interest compounding **periods** are known.
- Present value and future value **apply to interest calculations** on both **single payment** amounts and **periodic equal** payment amounts (**annuities**)

5.3.1. Future value of a Single Sum

- The accumulated amount of a single amount invested at compound interest may be computed period by period by a series of multiplication, as illustrated above for Br. 10,000 invested for one year at 8% compounded quarterly.
- If **n** is used to represent the number of periods that interest is to be compounded, **i** is used to represent the interest per period, and **p** is the principal amount invested, the series of multiplications to compute the accumulated amount (**FV**) in the example above determined as flows:

$$FV = p (1 + i)^n$$

$$FV = \text{Br. } 10,000 (1.02)^4$$

$$FV = \text{Br. } 10,000 [(1.02) (1.02) (1.02) (1.02)]$$

$$FV = \text{Br. } \underline{\underline{10,824.32}}$$

$$i = j/m = 8\%/4 = 2\% \\ n = m \cdot t = 1 \cdot 4 = 4$$

Exercise:

1. Find the future value of a \$10,000 investment at 2% annual interest compounded semiannually for three year.
2. A loan of \$2,950 at 8% is made for two years compounded annually. Find the future value (compound amount) of the loan. Find the amount of interest paid on the loan.
3. Davis invested \$20,000 that earns 6% compounded monthly for four years. Find the future value of Davis's investment.

Illustration of computation of future amount

- If on the day her daughter was born, Bethel deposited Br. 10,000 in a savings account that guarantees to accumulate interest quarterly at 10% a year.
- What will be the amount in the savings account on her daughter's 18th birthday?
- **Solution:** The amount in the savings account on her daughter's 18th birthday will be Br. 10,000 $(1 + 0.025)^{72}$. Because Table does not go beyond 50 periods, the amount in the savings account on the daughter's 18th birthday may be computed as follows:

$$\begin{aligned} \text{FV} &= \text{Br. } 10,000 (1 + 0.025)^{50} \times (1 + 0.025)^{22} \\ \text{FV} &= \text{Br. } 10,000 (3.437109) \times (1.721571) \\ &= \underline{\underline{\text{Br. } 59,172}} \end{aligned}$$

$$\begin{aligned} i &= j/m = 10\%/4 = 2.5\% \\ n &= m \times t = 18 \times 4 = 72 \end{aligned}$$

Other Application of future value amount of single payment:

- The future value equation for single payment stated in this material can also be used to find interest rates, as well as, numbers of years that will be needed for the compounded amount to equal the desired value.

Finding the Interest rate:

- A useful approach is to treat the interest rate as an implicit interest rate and found by using the interest table (future value table of single payment).

- **To illustrate**, assume that you have invested 15,000 Birr today at a bank where it can grow to the future value of 17,900 Birr within three years from now into the future.
 - ✓ What is the **interest rate** that the bank should pay for your account in order to fulfill your desire?
- **To answer this question**, substituting those values into the future value of single payment equation, you get:

$$FV_3 = PV (1+i)^3$$

$$FV/PV = (1+i)^3$$

$$FVIF_{i,3} = (1+i)^3 = \frac{17,900}{15,000} = 1.193$$

- The future value **interest factor** in the interest (future value of single payment table) corresponding to the **unknown interest rate** (i) and a period of **3 years** $n = 3$) is **1.193**.
- Hence, look up the three year ($n=3$) row and read horizontally until you find the **table value** (future value interest factor) that is equal or the closest to the computed value of **1.193**.
- There is **no table value** that is exactly equal to 1.193.
- The **table value of 1.191** is found to be the **closest value** to 1.193 and it corresponds to **6 percent**.
- **Therefore**, the interest that the bank actually has to pay to your account is **slightly greater than 6 percent**.

*Finding the **number of years**:*

- The future (compound) value of single payment equation can be used to **estimate the number of years** that are **required** for a given amount of money deposited at a specific interest rate to produce or desired compound amount.
- **For example**, a deposit of 1000 Birr is made in an interest bearing account that pays 10 percent compounded yearly.
- Your goal as a depositor is to collect 1,500 Birr after an unknown number of years.
- **How many years** should you wait for the desired amount to be realized?

By substituting the values into the future value of single payment equation, you get:

$$FV_n = 1000 (1+I)^n$$

$$1,500 = 1000 (1+0.1)^n$$

$$= 1000 (1.1)^n$$

$$(1.1)^n = 1500/1000$$

$$= 1.5, \text{ by using logarithm}$$

$$n = \log 1.1 \ 1.5 = \log 1.5 / \log 1.1 = 0.176 / 0.041 = \underline{\underline{4.29 \text{ years}}}$$

- Again it is possible to look up the **10 percent column** in the future value interest factors (future value) table and **read vertically** until you find a table value that is **equal to 1.5** or closest to it.
- The closest table value is **1.611**, which corresponds to **five years (n=5)**.
- That means if the 1000 Birr is kept in the account that pays 10 percent for five years; the resulting compounding amount will be **1,611 Birr**.
- This **amount exceeds** the desired amount of 1,500 Birr. If the 1000 Birr is kept in the account **only for four years**, the table value is **1.464**. Hence, the 1000 Birr has to be kept in the account for a period **slightly greater than 4 years**.

5.3.2. Present Value of a Single Sum

- Many measurement and valuation problems in financial accounting require the computation of the discounted present value of a principal amount to be paid or received on a fixed date.
- The present value represents the discounted amount (interest excluded) that will accumulate to the future amount (interest included).
- The present value of a future amount is always less than that future amount.
- The computation of the present value of a single future amount is a reversal of the process of finding the amount to which a present amount will accumulate.

FV = **p** (1 + **i**)^{**n**}, and when we solve for **p** by dividing both sides of the equation by (1 + **i**)^{**n**}, we have

$$\mathbf{PV} = \frac{FV}{(1 + i)^n}$$

Therefore, the formula for the **PV** of **FV** due in **n** periods at **i** rate of interest per period is

$$\mathbf{PV} = \frac{FV}{(1 + i)^n}$$

FV = future amount

P = present value

i = interest rate per period

n = number of compounding period

Example: If we want an amount of Br. 30,000 after 12 years by making a **single deposit** in a saving account which will pay 16% interest compounded **quarterly**, what should the amount of **initial deposit** be?

Solution: The present value is Br. 30,000 discounted at 4% for 48 periods.

The present value is

$$PV = \frac{FV}{(1+i)^n}$$

$$i = j/m = 16\%/4 = 4\%$$

$$n = m \cdot t = 12 \cdot 4 = 48$$

$$= \frac{Br.30,000}{(1 + 0.04)^{48}} = \frac{Br.30,000}{6.570528}$$

$$= \text{Br. } \underline{4565.84}$$

Exercise:

Abnet needs \$20,000 in five years to buy a new machine. How much must he invest at the present if he receives 5% interest compounded semi-annual?

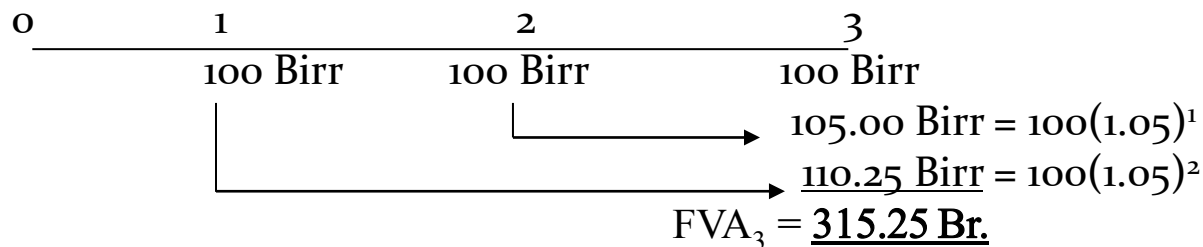
Annuities

- An annuity is a series of uniform payments or receipts occurring at uniform intervals over a specified investment time frame, with all amounts earning compound interest at the same rate.
- Payments of any type are considered as annuities if all of the following conditions are present:
 - i. The periodic payments are equal in amount
 - ii. The time between payments is constant such as a year, half a year, a quarter of a year, a month etc.
 - iii. The interest rate per period remains constant.
 - iv. The interest is compounded at the end of every time.

- Annuities are **classified** according to the **time the payment is made**.
- Accordingly, we have **two basic types of annuities**.
 - 1) **Ordinary annuity**: is a **series** of equal periodic **payment is made** at the **end of each** interval or period.
 - ✓ In this case, the **last payment** **does not earn** interest.
 - 2) **Annuity due**: is a type of annuity for which a **payment is made** at the **beginning of each** interval or period.
- ❖ **Amount of Annuity**: Amount of an annuity is **future values** of a **series of equal** receipts or payments (rents) made at **regular time intervals** and at the **same rate of interest** compounded each time the receipts or payments are made.

5.4.1. Future Value of an ordinary Annuity

- An **ordinary** consists of a series of equal payment made at the end of each period.
- If you deposit 100 Birr at the end of each year for three years in a saving account that pays 5 percent per year, how much will you have at the end of year three ($n=3$)?
- To answer this question, you must find future value of an annuity (FVA_n .)
- Each payment has to be compounded out to the end of period n , and the sum of the compounded payments gives you FVA_n .
- This can be shown by using the following time line.



$$FVA_n = PMT(1+i)^0 + PMT(1+i)^1 + PMT(1+i)^2 + \dots + PMT(1+i)^{n-1}$$

- In this equation the first term (i.e. $PMT(1+i)^0$) is the compounded value of the payment at the **end of last year**, or year **n** of the annuity payments; **while** the last term in the equation (i.e. $PMT(1+i)^{n-1}$) is the compounded amount of the payment made at the **end of year one (n-1)**.
- The above equation can further be simplified to:

$$FVA_n = (PMT) \left[\frac{(1+i)^n - 1}{i} \right] \quad \text{PMT} = \text{Per period payment amount}$$

- Using this future value of an annuity equation, the **future value** of the 100 Birr deposits made at the **end of each year** for three years at an interest rate of 5 percent would be:

$$\begin{aligned} FVA_3 &= (100) \left[\frac{(1+0.05)^3 - 1}{0.05} \right] \\ &= (100) (3.1525) = \underline{\underline{315.25 \text{ Birr}}} \end{aligned}$$

Example: Compute the amount of an **ordinary annuity** of 16 rents of Br. 100 at 2%

Solution:

$$FVAn = Br. 100 \left[\frac{(1+0.02)^{16} - 1}{0.02} \right] = \left(\frac{1.372786 - 1}{0.02} \right)$$

$$100 \left(\frac{0.372786}{0.02} \right)$$

$$= Br. 1863.93$$

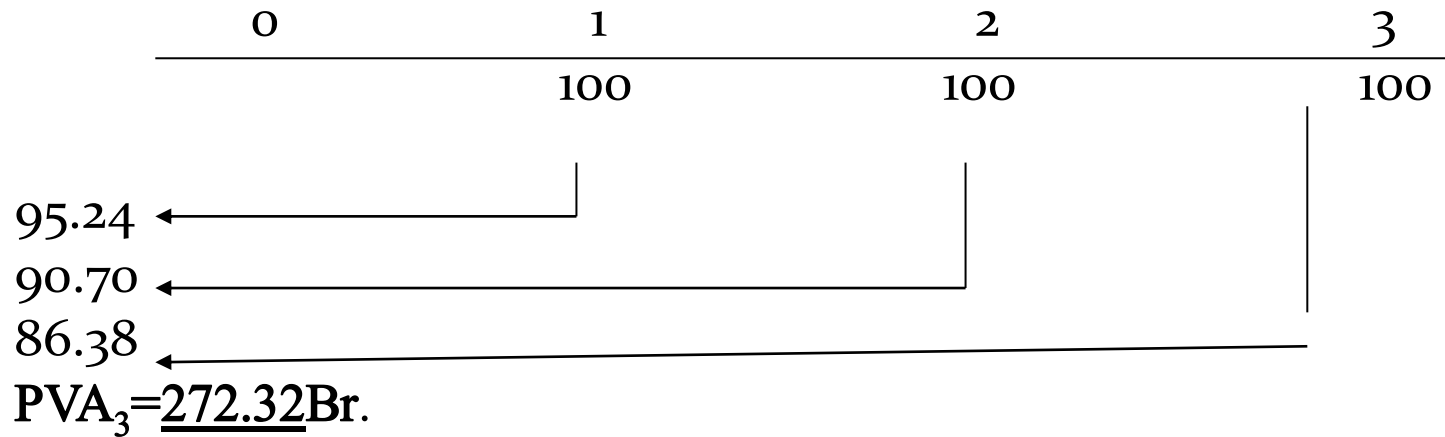
Exercise:

1. Find the value of an ordinary annuity after two years of \$1,500 invested semiannually at 4% annual interest?

5.4.2. **Present** (Discounted) value of **ordinary Annuity**:

- Suppose that you are offered **two alternatives**:
 - (1) a three-**annuity** with payment of **100 Br.** at the end of each year over the coming **three years**, or (2) a: lump-sum payment today.
- 1) You have **no need** for the money during the next three years, so **if you accept the annuity**, you would simply **deposit the payments** in the saving account at Dashen Bank that pays **5 percent** interest rate **compounded yearly**.
- 2) Similarly the **Lump-Sum payment** would be **deposited** in the same account as you don't have other option.
- How large should the Lump-sum payment today in order for it **be equivalent** to the annuity?

To answer this question, let us start-up with the aid of the **time line**.



- The present value of the series of payments of 100 Birr for three years annuity, PVA_3 is **272.32 Birr** as shown with the help of the time line.
- The same problem can be expressed by using **mathematical equation**.
- The **general mathematical equation** that can be **used to find** the **present value of an ordinary annuity** is shown below:

$$PVA_n = (PMT) \left(\frac{1}{1+i} \right)^1 + (PMT) \left(\frac{1}{1+i} \right)^2 + \dots + PMT \left(\frac{1}{1+i} \right)^n$$

Since PMT is **common for all terms**, the above equation can be **re-written** as:

$$PVA_n = (PMT) \left(\frac{1}{1+i} \right)^1 + \left[\left(\frac{1}{1+i} \right)^1 + \left(\frac{1}{1+i} \right)^2 + \dots + \left(\frac{1}{1+i} \right)^n \right]$$

Again the equation can be **rewritten** as:

$$PVA_n = (PMT) \left[\sum_{t=1}^n \left(\frac{1}{1+i} \right)^t \right]$$

The **summation term** in this equation is called the **present value interest factor** for an annuity (PVIFA) and it equivalent to: $\frac{1 - \frac{1}{(1+i)^n}}{i}$, or $\frac{1 - (1+i)^{-n}}{i}$

$$PVA_n = (PMT) \left[\frac{1 - (1+i)^{-n}}{i} \right], \text{ or } PMT \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right]$$

- By using this **mathematical equation**, the **present value of an annuity** of the problem under consideration can be computed as follows.
- You are given an annuity payment, PMT of 100 Birr, interest rate, 5 percent compounded yearly and an annuity period, **n** of three years.
- Substituting these values into the above equation, you get the present value of an annuity of **272.32 Birr**, which was the **same as the amount computed by summing the individual discounted valued** on the time line.

$$\begin{aligned}
 PVA_3 &= (100) \left[\frac{1 - (1 + 0.05)^{-3}}{0.05} \right], \text{ or } (100) \left[\frac{1 - \frac{1}{(1 + 0.05)^3}}{0.05} \right] \\
 &= (100) (2.7232) \\
 &= \underline{\underline{272.32 \text{ Birr}}}
 \end{aligned}$$

5.4.3. Future value of Annuity Due

- The amount of an annuity due (or annuity **in advance**) is the total amount on deposit/payment **one period after** the **final** deposit/payment.
- Similar to ordinary annuities **except** that payment/deposits are made **at the beginning** of each deposit/payment periods.

$$FVA (DUE)_n = PMT \left[\left\{ \frac{(1+k)^n - 1}{k} \right\} \times (1+k) \right].$$

Example. If the cooperative unions deposit 100 Birr interest of 10% for 3 years

Solution;

$$\begin{aligned} FVA (Due)_3 &= PMT \left[\left\{ \frac{(1+k)^3 - 1}{k} \right\} X(1+k) \right] \\ &= 100 \left[\left\{ \frac{(1+0.1)^3 - 1}{0.1} \right\} X(1+0.1) \right] \\ &= 100 [3.31]X(1.1) \\ &= \underline{\underline{Br. 364.1}} \end{aligned}$$

5.4.4 Present value of Ordinary Annuity

- Present value of ordinary annuity is the **discounted value** of a series of future rents on a date **one period before** the **first** payment/deposit.

$$PVA(DUE)_n = PMT \left[\left\{ \frac{1 - \frac{1}{(1+k)^n}}{k} \right\} \times (1+k) \right]$$

- You are given an annuity payment, PMT of **100 Birr**, interest rate, **10 percent** compounded yearly and an annuity period, n of **three years**. Calculate the PVA (DUE)_n?

Solution: $PVA(DUE)_n = PMT \left[\left\{ \frac{1 - \frac{1}{(1+k)^n}}{k} \right\} \times (1+k) \right]$

$$PVA(DUE)_n = 100 \left[\left\{ \frac{1 - \frac{1}{(1+0.1)^3}}{0.1} \right\} \times (1+0.1) \right]$$

$$PVA(DUE)_n = 100 \left[\left\{ \frac{1 - \frac{1}{(1+0.1)^3}}{0.1} \right\} \times (1+0.1) \right]$$

$$PVA(DUE)_n = \underline{\underline{\text{Br. 273.55}}}$$

5.4.5. Present value of **Perpetual Annuity**

- When an **annuity** is expected to **occur indefinitely**, it is called perpetual annuity.
- **Perpetuity** is a **stream of equal payments** expected to **continue forever**.
- The **present value** of perpetuity is:

$$PVP = \frac{\text{Payment}}{\text{Interest rate}} = \frac{PMT}{k}$$

For example, if the interest rate were **12 percent**, perpetuity of **\$1,000** a year would have a **present value** of = **\$8,333.33**.

$$\begin{aligned} PV_p &= \text{Payment/Interest rate} \\ &= PMT/k \\ &= \$1,000/0.12 \\ &= \underline{\underline{\text{Br. } \$8,333.33}} \end{aligned}$$

Exercise:

1. What is the future value of an annuity due with an annual payment of \$1,000 for three years at 4% annual interest? Find the total investment and the total interest earned.
2. If you make six monthly payments of \$50 to an annuity due and receive 6% annual interest compounded monthly, how much will you accumulate?



End of Chapter – Three

Thank You!!!



Chapter Four

Risk and return

RISK AND RETURN

- Risk and return fundamentals
- To maximize share price, the financial manager should assess two key determinants, risk and return.
- **Risk:** the chance of financial loss or more formally, the variability of return associated with a given asset – uncertainty.
- **Return:** the **total gain or loss** experienced on an investment over a given period of time.

Cont'd....

- The **return** is the basic motivating force and the principal **reward** in the investment process.
- Return can be defined in terms of
 - I) realized return
 - II) expected return
- **Realized return** is the return which **has been earned**.
- **Expected return** is the return which the investor anticipates to earn over some future investment period.

The realized returns in the past allow an investor to estimate cash inflows in terms of **dividends, interest, bonus, capital gains**, etc, available to the holder of the investment.

Cont'd...

- **Return** is commonly measured as a **cash distribution** during a period plus the **change in value**.
- The equation for calculating the **rate of return** earned on any asset over a period of time t is:

$$\text{Rate of return} = \frac{C_t + [P_t - P_{t-1}]}{P_{t-1}}$$

K_t = actual, expected, or required rate of return during period t ; C_t = cash flow received from the asset investment in the time period $t-1$ to t ; P_t = price (value) of asset at time t ; P_{t-1} = price (value) of asset at time $t - 1$.

Cont'd...

- Rate of return = $\frac{I_t}{P_{t-1}} + \frac{P_t - P_{t-1}}{P_{t-1}}$
- Where as $\frac{I_t}{P_{t-1}}$ is called **current yield**
- $\frac{P_t - P_{t-1}}{P_{t-1}}$ is called **capital Gain**
- **RoR = Current yield + Capital gain yield**
- **Example:** The following information is given for a corporate bond. Price of the bond at the beginning of the year: Br. 90, Price of the bond at the end of the year: Br. 95.40, Interest received for the year: Br. 13.50. Compute the rate of return.
- Calculate: current yield & capital gain yield

Cont'd...

Example:

ABC Co. wishes to determine the return on two of its video machines, A and B. A was purchased one year ago for \$20,000 and currently has a market value of \$21,500. During the year, it generated \$800 of after tax cash receipts. B was purchased 4 years ago; its value in the year just ended declined from \$12,000 to \$11,800. During the year it generated \$1,700 after tax cash receipts.

- Calculate the annual rate of return for each video machine.

Cont'd..

- **Solution:**

- Return (K_t) on A = $\frac{800 + 21,500 - 20,000}{20,000} = 11.5\%$

- Return (K_t) on B = $\frac{1,700 + 11,800 - 12,000}{12,000} = 12.5\%$

Video B's return is better than video A's return.

Cont'd....

- **Exercise:**
- The average market prices and dividend per share of Blue Ltd. for the past 6 years are given below:

Year	Average market price (Br.)	Dividend per share (Br.)
2002	38	1.8
2003	45	2.0
2004	53	2.5
2005	50	2.0
2006	61	2.6
2007	68	3.0

- Required: a) calculate each years yield rate?
- b) calculate each year's capital gain?
- c) calculate each year's rate of return?

- Determinants of the Rate of Return

- **Three** major determinants of the **rate of return** expected by the investor.

- The time preference risk-free real rate

- The expected rate of inflation

- The risk associated with the investment

- **Required return** = **Risk-free real rate** + **Inflation premium** + **Risk premium**.

- $RR = R_f + I_p + R_p$

RISK

- **Risk** refers to the chance that the **actual outcome** (return) from an investment will differ from **an expected outcome**.
- The risk may be considered as a chance of **variation** in return.
- Investments having greater chances of **variations** are considered **more risky** than those with **lesser** chances of variations.
- The absolute measures of risk is **standard deviation**
- **Type of risk:**
 - I) systematic risk/non-diversifiable
 - II) unsystematic risk/ diversifiable

Systematic Risk/ Non-diversifiable risk

- It refers to fluctuation in return due to **general** factors in the **market** such as **money supply**, **inflation**, **economic** situation, **interest rate** policy of the **government**, **political** factors, **credit** policy, **tax** reforms, etc. these are the factors which affect almost all firms.
- The systematic risk is also called the **general risk**.

Types of Systematic Risk

- **Market risk**: is refers to variability in return due to change in **market price** of investment. E.g. social, Economic and political situation.
- **Interest Rate Risk**: is refers to the variability in return caused by the change in level of interest rates.
- **Purchasing-Power Risk/ Inflation Risk**: it refers to the uncertainty of purchasing power of cash flows to be **received out of investment**. It shows the impact of **inflation** or **deflation** on the investment.

Unsystematic Risk/ diversifiable risk

- The unsystematic risk represents the **fluctuation** in return from an investment due to factors which are **specific to the particular firm** and not the market as a whole.
- these factors are **unique** to a particular firm, these factors called firm **specific factor**
- Types of Unsystematic Risk:
 - Business risk: the variability in incomes of the firms
 - Financial risk: the degree of leverage or degree of debt financing used by a firm
 - Operational risk: It occurs due to **breakdowns** in the internal procedures, people, policies and systems.

Risk and Expected Return

- Risk and expected return are the two key determinants of an investment decision.
- Risk is the variability of the rates of return from an investment.
- It measures how much individual outcomes deviate from the expected value.
- Statistically, risk is measured by any one of the measures of dispersion such as variance, standard deviation.
- Another major factor determining the investment decision is the rate of return expected by the investor.
- The rate of return expected by the investor consists of the yield and capital appreciation.

Risk measurement

□ the measure of risk is the extent to which the actual outcome is likely to diverge from the expected return. For this purpose, **range, variance, standard deviation and coefficient of Variation** can be used to measure the risk.

□ **standard deviation** is the absolute measure of risk

- **Example:** The rate of return of equity shares of Wipro Ltd., for past six years are given below:

Year	1	2	3	4	5	06
Rate of return (%)	12	18	-6	20	22	24

- Required: Calculate average rate of return, variance and standard deviation

Cont'd...

- Example 3: expected return and estimated risk
- Expected Return($E(R)$ or $\bar{R}$)= $\sum R \cdot P$
- Variance: $\delta^2 = \sum (R - \bar{R})^2 \times P$
- Standard deviation: $\delta = \sqrt{\delta^2} = \sqrt{\sum (R - \bar{R})^2 \times P}$
- Mr. Red invested in equity shares of White Ltd., its anticipated returns and associated probabilities are given below:

Return (%)	-15	-10	5	10	15	20	30
Probability	0.05	0.10	0.15	0.25	0.30	0.10	0.05

- Required: Calculation of expected rate of return and risk in terms of standard deviation?

cont'd...

Return(R)	Probability (P)	(P × R)	(R- $\bar{R}$)	(R- $\bar{R}$) ²	(R- $\bar{R}$) ² × p
-15	0.05	- 0.75	-24.5	600.25	30.0125
-10	0.10	-1.0	-19.5	380.25	38.0250
5	0.15	0.75	-4.5	20.25	3.0375
10	0.25	2.50	0.5	0.25	0.625
15	0.30	4.50	5.5	30.25	9.0750
20	0.10	2.00	10.5	110.25	11.0250
30	0.05	1.50	20.5	420.25	21.0125
	1.00	$\bar{R}$ = 9.5%		$\Sigma(R-\bar{R})^2 \times P = 112.8125$	

Cont'd...

- Example:
- Suppose have estimated possible returns for New Honda Dealership Inc. and Harry's Automotive Repair for the coming year based on how the economy does
- | | Return on: | | |
|----------------|--------------|-------------------|---------------|
| <u>Economy</u> | <u>Prob.</u> | <u>Dealership</u> | <u>Repair</u> |
| • Boom | 0.25 | 40% | 6% |
| • Average | 0.55 | 15% | 15% |
| • Bust | 0.20 | -1% | 17% |
- You are required to calculate the expected rate of return and risk in terms of standard deviation.

Cont'd...

- **Solution:**
- $E(r) \text{ Dealership} = 0.25(40) + 0.55(15) + 0.2(-1) = 18.05\%$
- $E(r) \text{ Repair} = 13.15\%$
- $\sigma^2 \text{ dealership} = 0.25(40 - 18.25)^2 + 0.55(15 - 18.25)^2 + 0.22(-1 - 18.25)^2 = 198.1475$
- $\sigma \text{ dealership} = 14.08\%$
- $\sigma \text{ Repair} = 4.20\%$
- Return is more uncertain for the dealership

Portfolio theory

- **Portfolio** is the combination of various stocks in it.
- Portfolio Management basically deals with **three** critical questions of investment planning.
 - 1. Where to Invest?
 - 2. When to Invest?
 - 3. How much to Invest?
- **Portfolio is the combination of assets**

Cont'd...

- The **weights** are based on the **percentage composition** of the portfolio.
- The ultimate decisions to be made in investments are:
 - 1. What **securities** should be **held**?
 - 2. **How many Birr** should be **allocated** to **each**?

Diversification and portfolio risk

- **Diversification** allows investor to **reduce** the level of the **overall risk** of a portfolio by eliminating the impact of individual risk.
- Efficiently diversified portfolios are those which provide the **lowest possible risk** for any level of expected return.
- Principle of diversification states spreading an investment across a number of assets will eliminate some, but not all, of the risk.
Diversification is not putting all your eggs in one basket.

Cont'd..

- The process of **spreading** an investment **across assets** (and thereby forming a portfolio) is called **diversification**.
- The **principle of diversification** tells us that **spreading** an investment across many assets will **eliminate** some of the risk.
- Unsystematic risk is therefore can be easily **avoided** virtually at **no cost**.
- **Total risk = Systematic risk + unsystematic risk**
- **Types of Diversification**
- There are two types of diversification. These are:
 - ➡ Random or naive diversification and
 - ➡ Efficient diversification

Cont'd...

1. Random or naive diversification:

- It refers to the act of randomly diversifying without regard to relevant investment characteristics such as **expected return** and industry classification.

2. Efficient diversification

- **Efficient diversification** takes place in an efficient portfolio that has the **smallest portfolio risk** for a given level of expected return or the **largest expected return** for a **given level of risk**.

Portfolio Expected Return

Portfolio return = $\sum_{k=1}^n \left(\frac{\text{Dollar Amount of Asset } k}{\text{Dollar Amount of Portfolio}} \right) * \text{return on Asset } k$

$$E(R) = \sum_{i=0}^n W_r * R$$

- The dollar amount of an asset divided by the dollar amount of the portfolio is the **weighted average** of the asset and the sum of all weighted averages must equal 100%.

Cont'd...

- **Example** A portfolio consists of four securities with expected returns of 12%, 15%, 18%, and 20% respectively. The proportions of portfolio value invested in these securities are 0.2, 0.3, 0.3, and 0.20 respectively.
- **The expected return on the portfolio is:**
- **$E(R_p) = 0.2(12\%) + 0.3(15\%) + 0.3(18\%) + 0.2(20\%)$**
- **$= \underline{16.3\%}$**

PORTFOLIO RISK

- The **risk of a portfolio** is measured by the variance (or **standard deviation**) of its return.
- ⑩ To develop the equation for calculating portfolio risk we need information on **weighted individual security risks** and **weighted co-movements** between the returns of securities included in the portfolio.
- ⑩ **Co-movements** between the returns of securities are measured by **covariance** (an absolute measure) and **coefficient of correlation** (a relative measure).

Cont'd...

- $\text{Cov}(R_A, R_B) = \delta_{A,B} = \sum_{i=1}^N P_i (R_{Ai} - E[R_A])(R_{Bi} - E[R_B])$
- Where: $\sum_{i=1}^N (R_{Ai} - E[R_A])(R_{Bi} - E[R_B]) =$ the covariance between the returns on stocks A and B
 - N = the number of states
 - p_i = the probability of state i
 - R_{Ai} = the return on stock A in state i
 - $E[R_A]$ = the expected return on stock A
 - R_{Bi} = the return on stock B in state i
 - $E[R_B]$ = the expected return on stock B

Cont'd...

$$\text{Portfolio Variance} = W_A^2 \sigma^2 + W_B^2 \sigma^2 + 2 W_A W_B \text{Cov}(R_A, R_B)$$

- $S.D = \sqrt{[W_A^2 \sigma^2 + W_B^2 \sigma^2 + 2 W_A W_B \text{Cov}(R_A, R_B)]}$
- **Example:**
- A portfolio consists of two stocks. The value of stock A is \$60,000, and its standard deviation is 15%, while the value of stock B is \$90,000, and its standard deviation is 10%. There is a correlation of 0.85 between the two stocks.
- **Determine the variance.**

Cont'd..

- **Example:** suppose you have an opportunity of investing your wealth in asset X and asset Y. The possible outcomes of the assets in different status of economy are given below:

Status of economy	probability	return (%)	
		X	Y
A	0.1	-8	14
B	0.2	10	-4
C	0.4	8	6
D	0.2	5	15
E	0.1	-4	20

Cont'd..

- Expected rate of return for individual asset:

$$X = (.1)(-8) + (.2)(10) + (.4)(8) + (.2)(5) + (.1)(-4) = 5\%$$

$$Y = (.1)(14) + (.2)(-4) + (.4)(6) + (.2)(15) + (.1)(20) = 8\%$$

- Assume you decided to invest 50% of your wealth in X and 50% in Y. what is your expected rate of return on a portfolio consisting X and Y.

Cont'd...

- Solution:

Status of economy(1)	probability (2)	combined return(3)	expected return(2*3)
A	0.1	$.5 * -8 + .5 * 14 = 3$	0.3
B	0.2	$.5 * 10 + .5 * -4 = 3$	0.6
C	0.4	$.5 * 8 + .5 * 6 = 7$	2.8
D	0.2	$.5 * 5 + .5 * 15 = 10$	2.0
E	0.1	$.5 * -4 + .5 * 20 = 8$	<u>0.8</u>

Or $0.5 * 5\% + 0.5 * 8\% = 6.5\%$

The expected return of the portfolio is 6.5%

➤ Portfolio risk is determined by the magnitude and direction of the correlation of any two of the assets in the portfolio.

Cont'd...

- **Correlation:** is a statistical measure of the relationship b/n any two series of numbers.
- If the series move in the same direction, they are positively correlated. If the series move in opposite direction, they are negatively correlated.
- **Correlation coefficient(CC):** a measure of the degree of correlation b/n two series.

Cont'd..

- If $CC = +1$, the two series are perfectly +vely correlated.
- If $CC = -1$, the two series are perfectly –vely correlated. If $CC = 0$, no correlation.
- $CC(XY) = \frac{\text{covariance } XY}{(\text{st. deviation } X)(\text{st. deviation } Y)}$

Cont'd...

- Covariance of XY for the above example:

Status of economy	proba.	return		deviation from exp. Return		product of deviates. & probab.
		X	Y	X	Y	
A	0.1	-8	14	-13	6	-7.8
B	0.2	10	-4	5	-12	-12.0
C	0.4	8	6	3	-2	-2.4
D	0.2	5	15	0	7	0.0
E	0.1	-4	20	-9	12	-10.8
covariance =						-33

Cont'd...

- Standard deviation for X:

- **Variance (σ^2)** = $0.1(-8-5)^2 + 0.2(10-5)^2 + 0.4(8-5)^2 + 0.2(5-5)^2 + 0.1(-4-5)^2$
= 33.6%

- **St. deviation(σ)** = $\sqrt{33.6} = 5.8\%$

- Standard deviation for Y:

Variance (σ^2) = $0.1(14-8)^2 + 0.2(-4-8)^2 + 0.4(6-8)^2 + 0.2(15-8)^2 + 0.1(20-8)^2$
= 58.2%

- **St. deviation(σ)** = $\sqrt{58.2} = 7.63\%$

Cont'd...

- Therefore,
- Coefficient of correlation = $\frac{-33}{5.8\% * 7.63\%} = -0.746$
- Thus, security X and Y are negatively correlated.
- If an investor invests in the combination of these securities, he/she can reduce risk.

Exercise

Economic State	Probability of State	Asset A Return (%)	Asset B Return (%)
Boom	20%	22	6
Normal	55%	14	10
Recession	25%	7	12

Cont'd...

- **Required:**
 - 1. Calculate expected return of asset A & B
 - 2. Calculate variance of asset A & B
 - 3. Calculate covariance of Asset A & B
 - 4. Calculate portfolio variance(σ^2_P)
 - 5. Calculate portfolio standard deviation
 -

End of chapter four

thank you!!!

Chapter - Five

Bonds and Stock Valuation and Cost of Capital

➞ Bonds and Stock Valuation

- It must be noted that all classes of investors are interested in knowing the values of securities i.e. common stock, preference stock and bonds.
- Since, the investors would be influenced in their decision to buy/sell by two sets of values.
 - ✓ One his own value and
 - ✓ The value externally determined by the market and known as price.
- It is important to weigh the risk and return, which affect the valuation process of both the individual and whole collection of investors that constitute market.
- Therefore, the valuation is the key concept for investment decisions.
 - ✓ No buy-sell action will take place without values.

The Valuation Process

- The basic valuation process exercise rationality with cost, benefits, and uncertainty as important variables.
- The valuation process will be examined in view of;
 - ✓ The performance of a firm in relation to the performance of industry to which it belongs; and
 - ✓ The performance of the economy and the market in general.
- The three sequential steps in the valuations process would be as follows:
 - ✓ Economy analysis
 - ✓ Industry analysis
 - ✓ Company analysis

The General Valuation Frame Work

- Most investors look at price movements in security markets.
- They observe opportunities of capital gains in such movements.
 - ✓ All would wish if they could successfully predict them and ensure gains.
 - ✓ Few, however, recognize that value determines price and both change randomly.
 - ✓ It would be useful for an intelligent investor to be aware of this process.

The basic valuations model

- Value of a security is a fundamental variable depends on its promised return, risk and the discount rate.
- In fact the basic valuation model is present value procedure with the mention of fundamental factors like returns and discount rate.
- Given a risk adjusted discount rate and the future expected earnings flow of security in the form of interest, dividend, earnings, or cash flow, you can always determine the present value as follows.

$$PV = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$$

Where PV = Present value

CF = Cash flow (interest, dividend, earnings per time period) up to 'n' number of years

r = Risk adjusted discount rate

1. Valuation of Bonds

- Bonds represent **loans** extended by investors to corporations and/or the **government**.
- Bonds are issued by the borrower, and purchased by the lender.
- The **legal contract** underlying the loan is called a **bond indenture**.
- A bond is an **instrument** or **acknowledgement** issued by a business unit or government by **specifying** the **amount of loan**, **rate of interest** and the **terms of loan repayment**.
- When the bond is purchased, the **bond holder (owner)** **receives two things**:
 - i. **Interest payments**, which are a series of equal payments and
 - ii. Repayment of the **full principal** at its maturity, regardless of the price the bondholder paid for the bond.

In order **to value** a bond you **must understand** the following.

- **Par value.** It is the **amount** or **value** stated on the **face** of the bond.
 - ✓ It represent the **amount** of the **firm borrows** and **promises to repay** at the time of maturity.
- **Coupon Rate of Interest.** A bond carries a **specific interest rate**, called coupon rate.
 - ✓ The **interest payable** is simply **par value** multiplied by the **coupon rate**.
- **Maturity period.** Every bond will have maturity period.
 - ✓ On **completion** of the **maturity period** the **principal amount** has to be **repaid** as per the agreed terms while issuing such bonds call provision.
 - ✓ Sometimes bonds may **be issued under a provision** that the business will have **an option to pay back** the bond amount **before** the maturity period.
 - ✓ These are known as **callable bonds**.

- **The intrinsic value of a bond.** Is the **present value (PV)** of the **cash flow** stream (CF) **provided** to the **investor**, discounted at a required **rate** of return appropriate for the risk involved.
- The **coupon interest** payments and **principal** payments are **known** and the **present value** is determined by **discounting** these **future payments** from the issuer at an appropriate **discount rate** or market yield.

$$PV = \sum_{t=1}^n \frac{C}{(1+r)^t} + \frac{TV}{(1+r)^n}$$

$$V_{B(PVB)} = C \left[\frac{1 - \left(\frac{1}{(1+r)^n} \right)}{r} \right] + \frac{M}{(1+r)^n}$$

Where, PV = Present value of the bond today

C = Coupon rate of interest

TV = Terminal value repayable

r = Appropriate discount rate or market yield

n = Number of years to maturity

a) Valuation of **perpetual** bond; bond that never mature

$$V_{B(pv)} = \frac{C}{r}$$

Where; C= coupon rate of interest,
r = Appropriate discount rate or market yield

b) Valuation of **zero-coupon** bond

$$V_B = \frac{M}{(1+r)^n}$$

Where; M= Maturity Value
n = number of years

- **Example.** A 10% bond of Birr 1,000 issued with a maturity of five years at par. The discounted rate of marketing is 10%. The interest is paid annually. What would be the bond value?

Solution, the intrinsic value of bond is its NPV of cash flow

$$\begin{aligned}
 PV &= \frac{100}{(1 + .10)} + \frac{100}{(1 + .10)^2} + \frac{100}{(1 + .10)^3} + \frac{100}{(1 + .10)^4} + \frac{100 + 1000}{(1 + .10)^5} \\
 &= 100 \times 0.9091 + 100 \times 0.8264 + 100 \times 0.7513 + 100 \times 0.6830 + 1100 \times 0.620 \\
 &= 90.91 + 82.64 + 75.13 + 68.30 + 682.99 \\
 &= \underline{\underline{999.97}}
 \end{aligned}$$

Or;

$$\begin{aligned}
 V_{B(PVB)} &= 100 \left[\frac{1 - \left(\frac{1}{(1 + 10\%)^5} \right)}{10\%} \right] + \frac{M}{(1 + 10\%)^5} \\
 &= 100 \left[\frac{1 - \left(\frac{1}{(1.1)^5} \right)}{0.1} \right] + \frac{1000}{(1.1)^5} \\
 &= 100 \left[\frac{0.379}{0.1} \right] + 620.92 \\
 &= 100(3.79) + 620.92 \\
 &= \underline{\underline{999.97}}
 \end{aligned}$$

Class work, A bond of Birr 1,000 at 6% is issued at par. The bond had a maturity period of five years. As of today five more years are left for final repayment at par. The current discount rate is 10 percent. What is the present value?

2. Valuation of Preferred Stock

- Preference shares are **hybrid security**. They have some features of bonds and some of equity shares.
- Theoretically, preference shares are **considered a perpetual security** but there are convertible, callable, redeemable and other similar features, which enable issuers to terminate them within the finite time horizon.
- Preference dividends are specified like bonds.
- Claims of preferred stockholders are junior to claims of debt holders, but senior to those of common stockholders.
- Preferred share holders have limited voting rights compared to common stock.
- Preferred stock has a par value and a dividend rate.

- Preference shares are **less risky** than **equity** because their **dividends** are **fixed** and **all arrears** must be **paid before equity** holders get their dividends.
- They are however, **more risky** than **bonds** because they **enjoy second priority** in **repayment** and in **liquidation**.
- Investor's **required returns** on preference shares are **more than** those on **bonds** but **less than** on **equity shares**.
- Since **dividends** from preference shares are **assumed** to be **perpetual payments**, the **intrinsic value** of such shares will be estimated from the following equations.

$$V_{ps} = \frac{D}{K_{ps}}$$

Where; V_{ps} = Value of preferred stock

D = Constant dividends received

K = required rate of return appropriate

- **Example.** A preference share of Birr 100 each with a specified dividend of Birr 11.5 per share. Now, if the investors' required rate of return corresponding to the risk level of a company is 10%, what would be the **value of share** today?

$$V_{ps} = \frac{D}{K_{ps}} = 11.5/0.10 = \text{Br } \underline{115}$$

- If the market price of the preference share is Birr 125 what would be the **yield**?

$$V_{ps} = \frac{D}{K_{ps}}$$

$$K_{ps} = \frac{D}{V_{ps}}$$

$$= 11.5/125 = \underline{9.2\%}$$

3. Valuation of Common Stock

- Common stock represents residual ownership of the firm.
- Common stockholders have important voting rights.
- The issuer may pay dividends to common stockholders.
- However, there is no pre-set dividend rate.
 - ✓ Future dividends are uncertain.
 - ✓ We need a way to forecast future dividends.
- In case of equity shares, the future stream of earnings or benefits poses two problems.
 1. One, it is neither specified nor perfectly known in advance as an obligation.
 - ✓ Resulting this, future benefits and their timing have both to be estimated in a probabilistic frame work.
 2. Two, there are at least three elements which are placed as alternative measures of such benefits namely dividends, cash flows and earnings.

- The valuation of common stock has **three methods**.
 - A. Zero growth model
 - B. Constant growth model
 - C. Super-normal growth model

A. Zero growth models

- Under this the assumption is the **growth of dividend** is **zero** or **constant**.

$$V_c = \frac{D}{K}$$

Where; V_c = Value of common stock

D = Dividend paid

K = the required rate of return

- **Example.** A company pays a cash dividend of Birr 9 per share on common share for an indefinite period of future. The required rate of return is 10% and the market price of the share is Birr 80.

✓ Would you **buy the share** at its current price?

$$V_s = \frac{D}{K} = 9/0.10$$

$$= \underline{\text{Br 90}}$$

Yes, you would consider **buying** the share.

B. Constant Growth Model

- The dividend payable to common stock holders will grow at a uniform rate in the future.
- It can be written as below.

$$V_c = \frac{D_0(1+g)}{k-g} = \frac{D_1}{k-g}$$

$$D_1 = D_0 + (D_0 \cdot g) = D_0 (1+g),$$

$$D_2 = D_0 (1+g)^2,$$

$$D_n = D_0 (1+g)^n, \text{ so}$$

$$V_{c_n} = D_0 (1+g)^n / (k-g)$$

Where D_0 = Dividend paid

g = growth rate

k = desired rate of return.

Example. Alfa Company paid a dividend of Birr 2 per share on common stock for the year ending March 31, 2003. A *constant growth of dividend 10% per share* has been forecast for an *indefinite future*. Investors *required rate of return is 15%*. You want to buy the share at market price quoted on July 31, 2003 is *stock market at Birr 60*.

- *What would be your decision?*

$$\begin{aligned}
 \text{Solution, } V_c &= \frac{D_0(1+g)}{k-g} \\
 &= \frac{2(1+10\%)}{(15-10)\%} = \frac{2(1.1)}{5\%} = \frac{2.20}{0.05} \\
 &= \underline{\text{Br 44}}
 \end{aligned}$$

Decision; Value is less than price, so you *do not buy*.

Class work; Nissan Ltd paid a dividend of Birr 4 per share for the ending march 31, 2003. The growth rate is 10% forever. The required rate of return is 15%. You want to buy the share at a market price of Birr 80 in stock exchange.

- What would you do?

C. Supper-normal Growth Model

- The **multiple growth** assumption has to be made in a vast number of practical situations.
- The **infinite future time period** is viewed as **divisible** into **two** or **more different** segments.
- The investor must **forecast** the time 'T' up to which growth would be variable and after which **only the growth rate** would show a pattern and would be constant.
- This mean that **present value calculations** will have to be **spread over two phases** viz. **one phase** would last until time 'T' and the **other** would **begin** after 'T' to maturity.

$$V_{T(1)} = \frac{D_t}{(1+k)^t}$$

$$V_{T(2)} = \frac{D_T + 1}{(k-g)(1+k)^T}$$

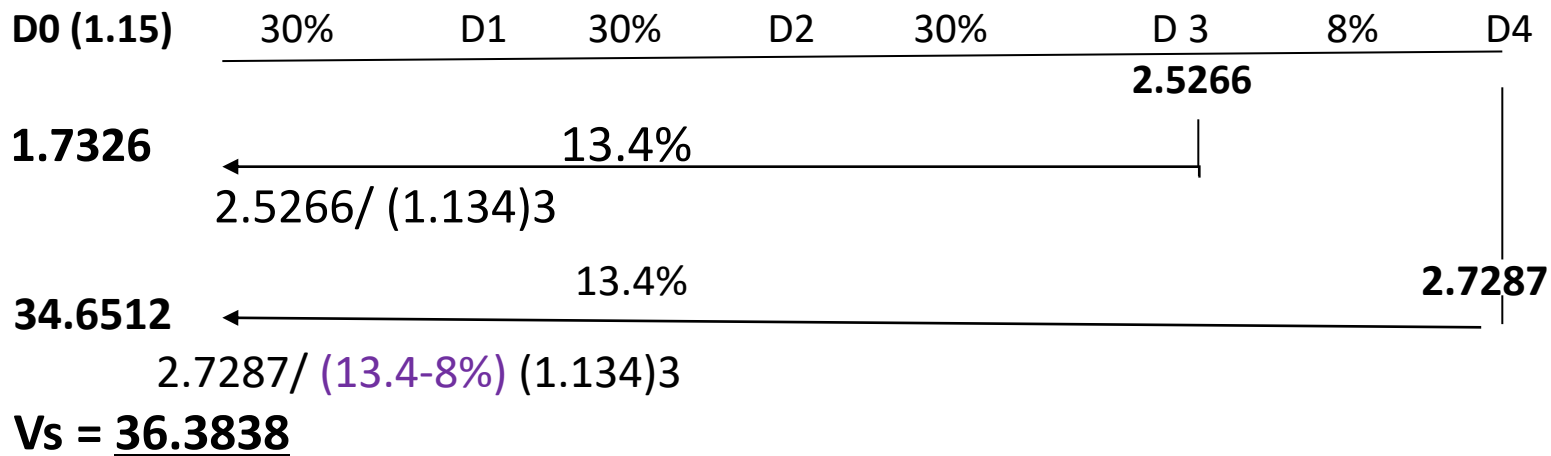
Combined equation for $V_{T(1)} + V_{T(2)}$

$$= \frac{D_t}{(1+k)^t} + \frac{D_T + 1}{(k-g)(1+k)^T}$$

- **Exmple.** Ethio-Power Corporation paid dividend of 1.15 Br per share during the last year. At this time, the forecast is that dividends will grow at 30% for the next three years and by 8% for the fourth year. The required rate of return is 13.4% for the next four years.

Solution: This is a case of multiple growths.

$$\begin{aligned}
 V_s &= \frac{D_t}{(1+k)^t} + \frac{D_{t+1}}{(k-g)(1+k)^T} & D_3 &= 1.15(1.30)^3 = 2.5266 \\
 & & D_{t+1} &= 2.5266(1.08)^3 = 2.7287 \\
 &= 2.5266 / (1.134)^3 & &= 1.7326 \\
 &= 2.7287 / (13.4\%-8\%) (1+13.4\%) & &= 34.6512 \\
 &= 1.7326 + 34.6512 & &= \underline{36.3838}
 \end{aligned}$$



➔ Cost of capital

What is cost of capital?

- 🍏 Cost of capital is the **required rate of return** that a firm **must achieve** in order to **cover** the **cost of generating funds** in the marketplace.
- The firm **must earn a minimum of rate of return** to **cover** the **cost of generating funds** to finance investments; otherwise, **no one** will be **willing to buy** the firms **bonds, preferred stocks and common stocks**.
- Based on their **evaluation** of the **riskiness** of each firm, investors will **supply new funds** to a firm **only if it pays them the required rate of return** to compensate them for taking the risk of investing in the firms bonds and stocks.
- **If**, indeed, the cost of capital is the **required rate of return** that the firm **must pay to generate funds**, it becomes a **guideline** for measuring the **profitability** of different investments.

- 🍏 Another way to think of the cost of capital is as the **opportunity cost of funds**, for investors which to invest their funds in assets with the same risk as the firm.
- If the **firm does not** achieve the **return investors expect** (i.e. the investors opportunity cost), investors **will not invest** in the firms **debt** and **equity**.
 - ✓ As a result the **firms value** (both their debt and equity) will **decline**.

Cost of Specific Source of Capital

1. Cost of Debt

- It is the cost associated with raising one or more dollar by issuing debt.

$$K_d = K_i (1-T)$$

Where; K_d = after tax cost of debt
 K_i = before tax cost of debt
 T = tax rate

- **Example;** suppose the palm computer company can issue debt with a yield 6%. If the palms tax rate is 40%, what is the cost of debt after tax?

Solution;

$$K_d = K_i (1-T)$$
$$= 0.06 (1-0.4) = 0.0360 = \underline{3.6\%}$$

- The interpretation is that the firm must earn 3.6% on debt investment to satisfy the bondholders interest.
- ❖ **Class work;** a firm borrow at an interest rate of 11% and the federal marginal tax rate is 40%, calculate the after tax cost of debt?

2. Cost of Preferred Stock

- It is the **cost associated** with **raising** one or more dollar of **capital** by **issuing** shares of preferred stock.
- It is the **rate of return** that must be **earned** on the **preferred stock holder's investment** to satisfy their requirements.
- When the firm sells preferred stock it expects to **pay dividends** to investors in return for their money capital
- In order to express this **dividend cost** as a **yearly rate** the firm uses the **selling price** it receives after deducting whatever costs incurred in issuing the preferred stocks.

$$K_p = \frac{\text{Dividend}}{\text{Net Price of } P_s} = \frac{D_{ps}}{\text{price} - \text{flotation cost}} = \frac{D_{ps}}{NP_p}$$

- **Example;** a preferred stock selling for Br 500 with an annual stated dividend of Br 50 require a flotation cost of Br 10 per share. Determine the specific **cost of preferred stock** if the corporate tax rate is 40%?

Solution ; $K_p = \frac{D_{ps}}{NP_p} = \frac{Br\ 50}{Br\ 500 - 10} = \underline{10.20\%}$

The interpretation is that the **firm must earn 10.20% on preferred stock investment** to **satisfy** the preferred stockholders interest.

3. Cost of Common Stock

- The specific cost of common stock is a **minimum rate of return** that the firm must earn for its common stock holders in order to maintain the market value of the firm's equity.

$$K_c = \frac{D_o(1+g)}{NP} + g$$

- **Example;** Millie Company's common stock has **recent divided** per share of Br 12. It is found that the company dividend per share should continue to **increase at 6%** growth rate in to the indefinite future.
 - ✓ What is the **specific cost of the common stock** if a **market price of Br 100** with a **flotation cost** per share **Br 8** is expected up on selling the stocks?

Solution

Given; Np = market price-flotation cost

$$= \text{Br } 100 - 8 = \text{Br } 92/\text{share}$$

$$D_o = \text{Br } 12/\text{share}$$

$$g = 6\%$$

$$\begin{aligned} K_c &= \frac{D_o(1+g)}{NP} + g = \frac{12(1+6\%)}{92} + 6\% \\ &= 0.138 + 0.06 = \underline{\underline{19.8\%}} \end{aligned}$$

Activity

1. The Gentol Co. just issued a dividend of \$2.55 per share on its common stock. The company is expected to maintain a constant 5 percent growth rate in its dividends indefinitely. If the stock sells for \$43 a share, what is the company's cost of equity?
2. Kev Bank has an issue of preferred stock with a \$4.20 stated dividend that just sold for \$93 per share. What is the bank's cost of preferred stock?
3. ICU Window, Inc., is trying to determine its cost of debt. The firm has a debt issue outstanding with seven years to maturity that is quoted at 96 percent of face value. The issue makes semiannual payments and has an embedded cost of 4.9 percent annually. What is the company's pretax cost of debt? If the tax rate is 38 percent, what is the after-tax cost of debt?
4. Jiminy's Cricket Farm issued a 30-year, 6.3 percent semiannual bond 8 years ago. The bond currently sells for 107 percent of its face value. The company's tax rate is 35 percent. a. What is the pretax cost of debt? b. What is the aftertax cost of debt?

4. Weighted Average Cost of Capital

- In the previous cost computation we assume that firms finance its assets from only one source.
- A firm can also finance its assets by using different financing alternative available to it.
- It may use bonds, stocks or retained earnings.
- And the specific cost of capital will not answer the amount that should be earned from the asset which is financed from different sources.
- So what should we do then?
 - ✓ A good answer for this is to compute a composite rate which is an average for the different sources of financing.
 - ✓ We call this rate the weighted average cost of capital.
 - ✓ It is the composite of the individual cost of financing.
 - ✓ It is the function of the individual cost of capital and the percentage of funds provided by securities like debts, preferred stock, and common stock.

$$WACC = W_d * R_d (1-T) + W_{ps} * R_{ps} + W_c * R_c$$

Steps to determine WACC of the firm

- i. Calculate the **specific cost** of funds for all source of finance
 - ii. **Multiply** the **cost of specific** funds by the **proportion of cash funds**
 - iii. **Add** the product
- **Example;** Imagine that you went to finance your assets with 100,000 birr and you get this amount from the following different sources. From debt= Br 40,000 has specific cost of 8%, preferred stock = Br 20,000 has specific cost of 11%, and common stock 40,000 has specific cost of 18%.
- ✓ Compute the firms **WACC**?

Solution;

Source of finance	Amount	Specific cost of capital (1 st)	Proportion
Debt	40,000	8%	40000/100000 = 40%
Pref. stock	20,000	11%	20000/100000 = 20%
Comm. Stock	40,000	18%	40000/100000 = 40%
Total	100,000		100%

- 2nd step; multiply the cost of specific funds by the proportion of cash funds

$$\text{Debt} = W_d * R_d = 40\% * 8\% = 0.032$$

$$\text{Pref. stock} = W_p * R_p = 20\% * 11\% = 0.022$$

$$\text{Comm. Stock} = W_c * R_c = 40\% * 18\% = 0.072$$

- 3rd step; add the product

$$\text{WACC} = W_d * R_d + W_p * R_p + W_c * R_c$$

$$= 0.032 + 0.022 + 0.072$$

$$= 0.126$$

$$= \underline{12.6\%}$$

- The company should earn at least 12.6% annual return from its assets worth of 100,000 to satisfy capital providers interest.

Activity

1. Barger Corporation has a target capital structure of 75 percent common stock, 5 percent preferred stock, and 20 percent debt. Its cost of equity is 10 percent, the cost of preferred stock is 5 percent, and the pretax cost of debt is 6 percent. The relevant tax rate is 30 percent. a. What is the company's WACC?
2. Micro Spinoffs also has preferred stock outstanding. The stock pays a dividend of \$4 per share, and the stock sells for \$40. What is the cost of preferred stock?
3. Reactive Industries has the following capital structure. Its corporate tax rate is 35 percent. What is its WACC?

Security	Market Value	Required Rate of Return
Debt	\$20 million	8%
Preferred stock	\$10 million	10%
Common stock	\$50 million	15%

End of Chapter – Five

Thank You!!!



Learning Never Ends.....

CHAPTER SIX

Capital Budgeting and Investment Decisions

❑ Capital Budgeting

Introduction

- The word **Capital** refers to be the **total investment** of a company in money, tangible and intangible assets.
- Whereas **budgeting** defined by the “Rowland and William” it may be said to be the **art of building budgets**.
 - ✓ Budgets are a **blue print** of a **plan** and **action** expressed in **quantities**.

Definitions

- According to Charles T. Hrongreen,
 - ✓ Capital budgeting is **a long-term planning** for **making** and **financing** proposed capital out lays.
- Capital Budgeting is the **process** of making investment decisions in **long-term assets**.
- It is the **process** of deciding whether or not to invest in a particular project as all the investment possibilities may not be rewarding.

Need and Importance of Capital Budgeting

1. **Huge investments:** Capital budgeting requires huge investments of funds, but the available funds are limited, therefore the firm before investing projects, plan are control its capital expenditure.
2. **Long-term:** Capital expenditure is long-term or permanent in nature. Therefore financial risks involved in the investment decision are more. If higher risks are involved, it needs careful planning of capital budgeting.
3. **Irreversible:** The capital investment decisions are irreversible, are not changed back. Once the decision is taken for purchasing a permanent asset, it is very difficult to dispose off those assets without involving huge losses.
4. **Long-term effect:** Capital budgeting not only reduces the cost but also increases the revenue in long-term and will bring significant changes in the profit of the company by avoiding over or more investment or under investment.

Therefore before making the investment, it is required carefully planning and analysis of the project thoroughly.

Capital Budgeting Process

The following steps followed for capital budgeting, then the process may be easier;

- 1. Identification of various investments proposals:** The capital budgeting may have various investment proposals. The proposal for opportunities may be from the top management or may be even from the lower rank.
 - The heads of various departments analyze the various investment decisions, and will select proposals submitted to the planning committee of competent authority.
- 2. Screening or matching the proposals:** The planning committee will analyze the various proposals and screenings.
 - The selected proposals are considered with the available resources of the concern. This reduces the gap b/n the resources and the investment cost.
- 3. Evaluation:** After screening, the proposals are evaluated with the help of various methods, such as PBP, NPV, IRR

- The proposals are evaluated as Independent or Exclusive proposals.
 - ✓ Independent proposals are not compared with another proposal and may be accepted or rejected.
 - ✓ Mutually exclusive proposals are those which are compared with another proposals.
- 4. **Fixing property:** After the evaluation, the planning committee will predict which proposals will give more profit or economic consideration.
 - ✓ If the projects or proposals are not suitable for the concern's financial condition, the projects are rejected without considering other nature of the proposals.
- 5. **Final approval:** The planning committee approves the final proposals, with the help of the following:
 - (a) Profitability
 - (b) Economic components
 - (c) Financial viability
 - (d) Market conditions.
- The planning committee prepares the cost estimation and submits to the management.

6. **Implementing:** The **competent authority** **spends** the **money** and **implements** the proposals.
- ✓ While implementing the proposals, assign responsibilities to the proposals, assign responsibilities for completing it, within the time allotted and reduce the cost for this purpose.
7. **Performance review of feedback:** The final stage of capital budgeting is **actual results** compared with the **standard results**.
- ✓ The adverse or unfavorable results identified and removing the various difficulties of the project.
 - ✓ This is helpful for the future of the proposals.

Investment Decision Making

- Investment means an expenditure in cash or its equivalent during one or more time periods in anticipation of enjoying a net cash inflows in future time periods.
- One of the important functions of Financial Management is investment decisions.
- The success of a business unit depends upon the investment of resource in such a way that brings in benefits or best possible returns from any investment.
- If the financial resources were in abundance, it would be possible to accept several investment proposals which satisfy the norms of approval or acceptability.

- Since, **resources are limited**; a **choice has to be made** among the **various investment proposals** by evaluating their **comparative merit** to select the **relatively superior proposals** with the limited available resources.
- For this purpose, **we have to develop** some **evaluating techniques** for the appraisal of investment proposals.

Importance of Investment Decisions

- The **terms** in financial management like **investment decisions**, **investment projects**, and **investment proposals** are generally **associated with** application of long-term resources.
- What is **long-term**?
 - ✓ a period of **ten years and above** may be treated as long period.
- The **decisions** related to **long-term investment** is also known as **capital budgeting techniques**.

It is **important** because of the following **reasons**:

1. The investment decisions are the **vehicles** of a company **to reach** its **desired destiny**.
 - An appropriate decision would **yield fantastic results** whereas
 - A wrong decision
 - ✓ May **upset the whole financial plan** and endanger the **survival** of the **firm**.
 - ✓ Even the firm may be **forced** into **bankruptcy**.
2. Capital budgeting techniques **involve huge amounts** of **funds** and **imply permanent commitment**.
 - Once you invest in the form of fixed assets it is **not easy to reverse** the decision **unless you incur** heavy loss.
3. A capital expenditure decision has **its effect over a long period** of **time span** and company's **future cost**.
4. Investment decisions are **among** the firm's **most difficult decisions**.
 - The **cash flow uncertainty** is **caused** by economic, political, social and technological forces.

Types of Investment Proposals

The long-term funds are required for the following purposes:

- a) **Expansion:** To add capacity to its existing product lines or to expand existing operations.
 - For example, a manufacturing unit producing 100,000 units per year may intends to double the output by 200,000, this will obviously increase the need for funds for acquiring fixed and current assets.
- b) **Diversification:** To add new product line to the existing product lines.
 - Philips, a famous company for radio and electric bulbs etc. diversified into production of other electrical applications and television sets.

C) Replacements: Machines used in production may either wear out or may be obsolete on account of new technology.

- The productive capacity of the enterprise and its competitive ability may be adversely affected.
- Extra funds are required for modernization or renovation of the entire plant.

D) Research and Development: The efficiency of production and the total operations can be improved by application of new and more sophisticated techniques of production and management.

- To acquire the technology huge funds are needed.

The useful way of classifying investments is as under

1. Accept-Reject Decisions

- Under this, if a project is accepted, the firm is going to invest, otherwise it is not going to invest its funds.
- In general, the project proposals that yield a rate of return greater than the certain required rate of return or cost of capital are accepted and others are rejected.
- By applying this criteria more than one independent project are accepted subject to availability of funds.

2. Mutually Exclusive Decisions

- These are the projects which compete with each other in such a way that the acceptance of one will exclude/reject/ the other one.
- If the alternatives are mutually exclusive one may be chosen.

3. Capital Rationing

- We are aware that the financial resources are limited.
- But, a large number of investment proposals compete for those limited funds.
- Thus, capital rationing refers to the situation in which the firm has more acceptable investments, requiring a greater amount of finance than is available within the firm.
- Capital rationing refers to a situation where a firm is not in a position to invest in all profitable projects due to the constraints on availability of funds.
- It is for this reason the firm cannot take up all the projects though profitable, and has to select the combination of proposals that will yield the greatest profitability.
- The projects are ranked and selected as per their merits of acceptance basing on certain predetermined criteria.

Data required for investment decisions

- 🍏 **Initial Investment:** The total amount of cash required to buy various assets like land, buildings, plant, machinery, equipment, etc and there installation expenses have to be estimated.
 - ✓ In addition to fixed cost, the cost of maintaining stocks, contingency reserves to cover the cost of supporting the additional receivables.
- 🍏 **Subsequent Investment:** The cost of maintenance, replacement and updating are to be treated as outflows during the period expected to be incurred.
- 🍏 **Economic Life of the project:** The economic life of a project is to be distinguished from the life of an individual asset.
 - ✓ The economic life of the project is determined by the duration of the “earnings flow” generated by the project.

The economic life may **end**:

- ✓ The **cost of replacement** becomes **uneconomical** in relation to the likely **benefits**.
- ✓ When the **viability** is adversely affected due to **obsolescence**.
- ✓ When **maintenance costs** exceed the **disposable value**, and
- ✓ When the development of **new technology** need **new investment**

🍏 **Salvage Values:** The **plant assets** may **have value** for the enterprise **at the end** of the **life of the project** or there may be some **anticipated sales value** of the **plant**.

- ✓ Such amount is to be treated as **an inflow** at the end life of the project.

🍏 **Annual Cash Flows:** The calculation of annual cash flows in investment appraisal plays a key role.

❖ The computation of cash flows is a simple task. The following areas are to be considered.

➡ **Sales revenue:** This is going to be the function of sales.

✓ Care has to be taken to forecast accurately and the additional revenue generated by the investment should be taken into account.

✓ The investment must also result into the reduction of operating cost either by modernization or replacement models where the savings benefit or cash flows will increase.

➡ In simple terms annual cash flow is equivalent to net profit after tax plus depreciation. Procedurally,

$$\text{ACF} = \text{Sales} - (\text{Operating expenses} + \text{Non-operating expense}) + \text{Depreciation}$$

- **Example:** The following is the information related to ABC Ltd.

	<u>Machine A</u>	<u>Machine B</u>
	Birrs	Birrs
Cost of machine	15, 000	24, 000
Estimated life in years	5 years	6 years
Estimated income	1, 000	1, 500
Estimated material cost	800	900
Estimated supervision cost	1, 200	1, 600
Estimated maintenance cost	500	1, 000
Estimated savings in wages	9, 000	12, 000

Additional Information

- ✓ Depreciation may be charged on straight line method
- ✓ The tax rate may be assumed 50%

Required:

Calculate Annual Cash Flows?

	Machine A	Machine B
	<u>Birrs</u>	<u>Birrs</u>
Income:	1, 000	1, 500
Savings in wages	<u>9, 000</u>	<u>12, 000</u>
<i>Total income / sales</i>	<u>10, 000</u>	<u>13, 000</u>
Cost: Material costs	800	900
Maintenance	500	1, 000
Supervision	<u>1, 200</u>	<u>1, 600</u>
<i>Profit before Depreciation & Tax</i>	<u>7, 500</u>	<u>10, 000</u>
Depreciation	<u>3, 000</u>	<u>4, 000</u>
<i>Profit Before tax</i>	<u>4, 500</u>	<u>6, 000</u>
Tax @ 50%	<u>2, 250</u>	<u>3, 000</u>
<i>Profit after tax (Net profit)</i>	<u>2, 250</u>	<u>3, 000</u>
Add back depreciation	<u>3, 000</u>	<u>4, 000</u>
Annual cash flows	<u>5, 250</u>	<u>7,000</u>

Project Appraisal Methods

There are two important methods of evaluating the investment proposals

1. **Traditional methods**
 - A. Payback Period
 - B. Accounting Rate of Return
2. **Discounted cash flow method**
 - A. Net Present Value
 - B. Internal Rate of Return
 - C. Profitability Index

1. Traditional Methods

A. Payback Period:

- This is one of the **widely used methods** for **evaluating** the investment proposals.
- Under this method the **focus is on** the **recovery of original investment** at the **earliest possible** period.
- It determines the **number of years** to **get back** the **original cash out flow**, disregarding the salvage value and interest.
- This method **does not take into account** the **cash inflows** that are received after the payback period.

Decision rule: **accept** the **project** if it's **payback period** is **shorter than** the **target (planned period)** set by the management.

There are **two methods** in use to calculate the payback period

- 1) Where annual **cash flows** are **not consistent** (vary from year to year)
- 2) Where the annual **cash flows** are **uniform**

1) Unequal cash flows

$$P = E + \frac{B}{C}$$

Where, P = payback period.

E = number of years immediately preceding the year of final recovery.

B = the balance amount still to be recovered.

C = cash flow during the year of final recovery.

Example: The following is the information related to a company

	Project A	Project B
<u>Year</u>	<u>Cash flow \$</u>	<u>Cash flow \$</u>
0	-700	-700
1	100	400
2	200	300
3	300	200
4	400	100
5	500	0

Calculate payback period of two projects?

<u>Project A</u>		Cumulative
Year	Cash flow	cash flow
0	-700	-700
1	100	-600
2	200	-400
<u>3</u>	300	<u>-100</u>
4	<u>400</u>	300
5	500	800

<u>Project B</u>		Cumulative
Year	Cash flow	Cash flow
0	-700	-700
1	400	-300
2	300	0
3	200	200
4	100	300
5	0	-

Solutions;

$$\begin{aligned}
 P &= E + \frac{B}{C} \\
 &= 3 + \frac{100}{400} \\
 &= \underline{\underline{3.25 \text{ year}}}
 \end{aligned}$$

$$\begin{aligned}
 P &= E + \frac{B}{C} \\
 &= 2 + 0/200 \\
 &= \underline{\underline{2 \text{ years}}}
 \end{aligned}$$

2. Uniform cash flows:

- Where the **annual cash flows** are **uniform**

$$PB_p = \frac{\text{Original Investment}}{\text{Annual cash flows}}$$

Example: A project requires an investment of \$ 100, 000, it will generate annual cash flow of \$25,000 per year.

- Calculate the payback period?

$$\begin{aligned} PB_p &= \frac{\text{Original Investment}}{\text{Annual cash flows}} = \frac{100,000}{25,000} \\ &= \underline{\underline{4 \text{ years}}} \end{aligned}$$

B. Accounting Rate of Return (**keep on reading**)

- This method is based on the financial accounting practices of the company working out the **annual profits**.
- Here, **instead of taking** the **annual cash flows**, we take the **annual profits** into account.
- The **net annual profits** are calculated after **deducting depreciation** and **taxes**.
- The **average of annual profits** thus derived is worked out on the basis of the period

$$ARR = \frac{\text{Average annual profits after taxes}}{\text{Average investment over the life of project}} \times 100$$

2. Discounted Cash Flow Method

- This concept is **based on** the **time value of money**.
- The **real value** of **Birr** in your hand today is **better than** value of **birr** you earn after a year.
- The **future income**, therefore, has to be discounted in order to be associated with the **current out flow** of funds in the investment.
- Two methods of **appraisal** of **investment project** are based on this concept.
 - ✓ These are **NPV** and **IRR** method.

C. Net Present Value (NPV)

- It is an investment project proposals **evaluating** and **ranking method** **using** the **net present value**, which is the **difference** between the **present values of future cash inflows** and the **present value of cash outflows**, discounted at the given cost of capital, or opportunity cost of capital.

In order to use this method properly, the following procedures are followed.

1. **Find** the **present value** of **each cash flow** (both inflows and outflows) using the cost of capital of the project for discounting.
2. **Sum** the **discounted cash inflows** and the **discounted cash outflows** separately.
3. **Obtain** the **difference** between the **sum of the cash inflows** and the **sum of the cash outflows**.

- If all the cash outflows for the project occur at time zero, i.e. at the beginning of year 1, the present value of the cash out flows is the same as to the net investment amount.

Decision Rule; The decision rule here is; accept a project if the NPV is positive and reject if it is negative

⇒ NPV > Zero Accept

⇒ NPV = Zero Indifference

⇒ NPV < Zero Reject

- ⇒ If the projects are independent, the projects with positive net present values are the ones whose implementation maximizes the wealth of shareholders.

✓ Hence, such projects should be accepted for implementation.

- ➡ If the projects, on the other hand, are mutually exclusive, the one with the **higher positive NPV** should be **accepted** leading to the rejection of the projects with lower positive NPV.
- ➡ **NPV** of **zero** imply that the cash flows of the project are **just sufficient** to **repay** the **invested capital** and to provide the **required rate of return**, **no more, no less**.
- ➡ Projects with **negative NPV** should **not be considered** for acceptance in the **first place**.

$$NPV = \frac{CF_1}{(1+R)^1} + \frac{CF_2}{(1+R)^2} + \frac{CF_t}{(1+R)^t} - I_0$$

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+K)^t} - I_0$$

Where; NPV = Net present value

CF_t = Cash inflows at different periods

I₀ = Cash outflow in the beginning (initial investment)

K = Cost of capital

t = time period

Example; Assume that a given project is expected to have an initial investment and project life of 40,000 Birr and 5 years respectively. The annual after-tax cash flow is estimated at 12,000 Birr for each of the five years.

1. Using the required rate of return of 10 percent, what is the **net present value (NPV)** of the project?
2. How do you **judge** the acceptability of this project?
 - To answer these questions, it is wise to **identify** the **cash inflows** and **outflows**. In this project, there are **annuity** cash inflows of 12,000 every year for five years and **single cash out flow** of 40,000 at time zero.

- The **present value** of the **annuity cash inflows** is:

$$\text{Present value of annuity} = (12,000 * \text{Annuity factor})$$

- The **annuity factor** given the period of 5 years and discount rate of 10 percent is 3.791 **substituting** the factor I the equation above

$$\text{PVA} = (12,000) (3.791) = \underline{\underline{45,492 \text{ Birr}}}$$
 or

$$= [12000 / (1.1)] + [12000 / (1.1)^2] + \dots + [12000 / (1.1)^5] = 45,492 \text{ Birr}$$

$$\text{Present Value of Cash out flow} = \underline{\underline{40,000 \text{ Birr}}}$$

$$\text{Net Present value (NPV)} = \underline{\underline{5,492 \text{ Birr}}}$$

Hence,

The Net Present value (NPV) = Present Value of **inflows** less(-) present value of **outflows**

$$\text{NPV} = 45,492 - 40,000 = \underline{\underline{5,492 \text{ Birr}}}$$

- ✓ Since the project makes the net present value (**NPV**) of **positive** 5,492 Birr, the project should be **accepted**.
- ✓ Consequently, the **wealth of the shareholders** would **increase by 5,492 Birr** in total **as the result of accepting and running** this **project**.

To further illustrate the **NPV method**, consider the following **mutually exclusive project** alternatives, together with their cash flows.

Alternative	<u>Year 0</u>	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>	<u>Year 5</u>
A	(80,000)	20,000	25,000	25,000	30,000	20,000
B	(100,000)	25,000	20,000	30,000	35,000	40,000

- The required rate of return on both projects is **12 percent**.
- Then, **Evaluate** these projects using **Net Present Value method**?
 - ✓ The evaluation of these two projects **requires** the **computation** of the **net present values** for both projects.
 - ✓ **Thus**, we need to **discount each of the cash flows** individually.
 - ✓ **Then**, the individual discounted cash flows are **added**.
 - ✓ The **cash out flows** at time zero will be **deducted** from the sum of the **discounted cash inflows** in order **to get the net present value** of the project.

The net present value (NPV) for project A is:

Year	Cash flows	Discount Factor (12%)	Present Values
1	20,000	0.893	17,860
2	25,000	0.797	19,925
3	25,000	0.712	17,800
4	30,000	0.636	19,080
5	20,000	0.567	<u>11,340</u>
Present values of cash inflows (sum)			86,005
Present values of cash outflows			<u>80,000</u>
Net Present Value (NPV)			<u>6,005 Birr</u>

The net present value (NPV) for project B is:

Year	Cash flows	Discount Factor (12%)	Present Values
1	25,000	0.893	22,325
2	20,000	0.797	15,940
3	30,000	0.712	21,360
4	35,000	0.636	22,260
5	40,000	0.567	<u>22,680</u>
Present values of cash inflows (sum)			104,565
Present values of cash outflows			<u>100,000</u>
Net Present Value (NPV)			<u>4,565 Birr</u>

- Since the two projects are mutually exclusive, the one with the higher NPV has to be accepted.
- Thus, project A is selected as its NPV is higher than that of project B.
- Had the two project been independent of one another, both of them would be accepted because both projects have positive NPVs

D. Internal Rate of Return (IRR)

- The internal rate of return is the **discount rate** which **equates** the **present value of the expected cash flows** with the **initial investment outlays**.
- In other words, IRR is **a method** of **ranking investment project proposals** using the **rate of return** on an asset (investment).
- At IRR, the **sum** of the **present values of all cash inflows** is **equal** to the **sum** of the **present values of all cash outflows**.
- That is:

$$\text{PV (cash inflows)} = \text{PV (cash outflows)}.$$

- Hence, the **net present value of any project** at a **discount rate** that is **equal to the IRR** is **zero**.

$$I_0 = \sum CF_t / (1 + \text{IRR})^t$$

Decision rule for the IRR technique is;

If the $IRR \geq r$ (the cost of capital) = **Accept** project

$IRR < r$ (the cost of capital) = **Reject** the project

Example;

Year	0	1	2	3	4	5
After tax cash flow	- \$10,000	2,000	4,000	3,000	3,000	1,000

Required; Calculate **IRR** and decide whether to **accept or reject** the project, since the firm has a **cost of capital of 8%**?

Solution; in order to solve for IRR, we need to solve the following equation;

$$10,000 = \frac{2000}{(1+IRR)} + \frac{4000}{(1+IRR)^2} + \frac{3000}{(1+IRR)^3} + \frac{3000}{(1+IRR)^4} + \frac{1000}{(1+IRR)^5}$$

Assume $IRR = 10\%$; first trial and error

$$10,000 = \frac{2000}{(1+10\%)} + \frac{4000}{(1+10\%)^2} + \frac{3000}{(1+10\%)^3} + \frac{3000}{(1+10\%)^4} + \frac{1000}{(1+10\%)^5}$$

$$\begin{aligned}
 10,000 &= \frac{2000}{(1.1)} + \frac{4000}{(1.1)^2} + \frac{3000}{(1.1)^3} + \frac{3000}{(1.1)^4} + \frac{1000}{(1.1)^5} \\
 &= 1,818.18 + 3,305.79 + 2,253.94 + 2,049.04 + 620.92 \\
 &= 10,047.87
 \end{aligned}$$

IRR = 11%; second trial and error

$$\begin{aligned}
 10,000 &= \frac{2000}{(1.11)} + \frac{4000}{(1.11)^2} + \frac{3000}{(1.11)^3} + \frac{3000}{(1.11)^4} + \frac{1000}{(1.11)^5} \\
 &= 1,801.80 + 3,246.49 + 2,193.57 + 1,976.19 + 593.45 \\
 &= 9,811.5
 \end{aligned}$$

To find out the exact IRR, we can use interpolation system

$$\begin{aligned}
 \text{IRR} &= I_1 + (I_2 - I_1) \mid \frac{\text{NPV } I_1 - I_0}{\text{NPV } I_1 - \text{NPV } I_2} \mid \\
 &= 10 \% + (11\% - 10\%) \mid \frac{10,047.87 - 10,000}{10,047.87 - 9,811.5} \mid \\
 &= 10 \% + (1\% * 0.2025) \\
 &= 10\% + 0.002025 \\
 &= 0.10205 \\
 &= \underline{10.205\%}
 \end{aligned}$$

If the cash flow in **Annuity form**;

The following **steps** can be followed to calculate IRR for **constant cash inflows**.

Step 1: Find the critical value of **discount factor**

$$\text{Discount factor} = \frac{\text{Initial investment}}{\text{Annual Cash inflow}}$$

Step 2: Find the **IRR** by looking along the appropriate line (year) of the present value of **annuity table** until the column which contains the critical discount factor (i.e. the discount factor computed under step 1) is located.

- **Illustrate:-** Assume that a project has a net investment of 26,030 Birr and annual net cash inflows of 5000 Birr for seven years. What is the **IRR** of this project?

Step 1 Compute the critical discount factor. That is

$$\text{Discount factor} = \frac{26,030}{5,000} = 5.206$$

Step 2 look for the value that is equal to this discounting factor in the present value of annuity table across the line corresponding to **7 years** (i.e. $n=7$). The discount factor of **5.206** appears in the **8 percent column** on the line/row of 7 years. **Therefore, the IRR = 8 %.**

E. Profitability Index (PI)

- Profitability index is the **ratio** of the **present value** of the **expected net cash flow** of the project and its **initial investment outlay**.

$$PI = \sum \frac{CFT}{(1+r)^t} / I_0 = \frac{PV}{I_0} \quad \text{Where; } PV = \text{Present value of expected net cash flows}$$

$I_0 = \text{Initial investment outlay}$
 $PI = \text{Profitability Index}$

- Profitability index **measure** of **profitability** in a more readily understandable terms.
- It simply converts the NPV criterion into a relative measure.

*Consider the following two **mutually exclusive** projects.*

	<u><i>Present Value of cash Flow</i></u>	<u><i>Initial Investment</i></u>	<u><i>Profitability Index (PI)</i></u>
Project A	200	100	2.0
Project B	3000	2000	1.50

- From the above example **project A** is **accepted** using profitability index **because** its PI is **greater than** that of project B.

Methods of Capital Budgeting of Evaluation

- By **matching** the **available resources** and **projects** it can be invested.
- The funds available are always **existing** funds.
- There are many considerations taken for investment decision process such as **environment** and **economic** conditions.

The methods of evaluations are classified as follows:

(A) Traditional methods (or Non-discount methods)

- (i) Pay-back Period Methods
- (ii) Post Pay-back Methods
- (iii) Accounts Rate of Return

(B) Modern methods (or Discount methods)

- (i) Net Present Value Method
- (ii) Internal Rate of Return Method
- (iii) Profitability Index Method



End of Chapter – Six

Thank You!!!

